Team of One

Thawan

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1.1 binomial-inversion

```cpp
// Binomial Inversion: transforms subset group counts into exact
    frequency counts
// Time: O(n^2) | Space: O(n)
// Formula: exact[i] = sum{j=i}^{n} (-1)^(j-i) * C(j, i) * subsets[j]
vector<int> binomial_inversion(const vector<int> &subsets) {
    int n = subsets.size() - 1;
    vector<int> exact(n + 1);
    for (int i = 0; i <= n; ++i) {
        int count = 0;
        for (int j = i; j <= n; ++j) {
            int term = (C(j, i) * subsets[j]) % mod;
            if ((j - i) & 1) {
                count = (count - term + mod) % mod;
            } else {
                count = (count + term) % mod;
            }
        }
        exact[i] = count;
    }
    return exact;
}
```

1.2 catalan

```cpp
// Catalan (Ballot Theorem): number of grid paths from (0,0) to (l,r)
    staying x >= y at all times
// Time: O(1) | Space: O(1)
// Note: requires C(n, k) | also counts valid parenthesis prefixes (l
    open, r close)
int catalan(int l, int r) {
    if ((l < 0 or r < 0) or l < r) return 0;
    return (C(l + r, l) - C(l + r, l + 1) + mod) % mod;
}
```

1.3 lucas

```cpp
int lucas(int n, int k) {
    if (k == 0 or k == n) return 1;
    if (k > n) return 0;
    return (lucas(n / mod, k / mod) * C(n % mod, k % mod)) % mod;
}
```

1.4 permutation-combination

```
int P(int n, int k) {
    if (k < 0 or n < k) return 0;
    return fact[n] * invf[n - k] % mod;
}

int C(int n, int k) {
    if (k < 0 or n < k) return 0;
    return P(n, k) * invf[k] % mod;
}
```

1.5 stars-and-bars

```
// Stars and Bars: number of ways to put n identical items into k
    distinct bins
// Time: O(1) | Space: O(1)
// Note: requires C(n, k) function | counts non-negative integer
    solutions to x1 + ... + xk = n
int stars_and_bars(int n, int k) {
    if (n < 0 or k < 0) return 0;
    if (k == 0) return n == 0;
    return C(n + k - 1, k - 1);
}
```

2.1 bounded-subset-sum

```
// Bounded Subset Sum: maximizes subset sum without exceeding the
    target
// Time: O(N * max(W)) | Space: O(max(W))
// Note: efficient for huge targets with small weights
int bounded_subset_sum(vector<int> &w, int target) {
    int k = 0, s = 0;
    while (k < w.size() and s + w[k] <= target) s += w[k++];

    if (k == w.size() or s == target) return s;

    int bound = *max_element(w.begin(), w.end());
    int offset = target - bound;

    vector<int> prev(2 * bound + 1, -1);
    prev[s - offset] = k;

    for (int i = k; i < w.size(); ++i) {
        vector<int> curr = prev;

        // insert
        for (int x = 0; x < bound; ++x) if (prev[x] != -1) {
            curr[x + w[i]] = max(curr[x + w[i]], prev[x]);
        }

        // remove
        for (int x = 2 * bound; x > bound; --x) if (curr[x] >
            prev[x]) {
            for (int j = max(0ll, prev[x]); j < curr[x]; ++j) {
                curr[x - w[j]] = max(curr[x - w[j]], j);
            }
        }

        swap(prev, curr);
    }

    for (int x = bound; x >= 0; --x) if (prev[x] != -1) {
        return x + offset;
    }
    return 0;
}
```

2.2 divide-and-conquer

```
int divide_and_conquer() {
    vector<int> prev(n + 1, inf);
    vector<int> curr(n + 1, inf);
    auto compute = [&](auto &self, int l, int r, int opt_l, int
        opt_r) -> void {
        if (l > r) return;
        int m = (l + r) / 2;
        curr[m] = inf;
        int opt = opt_r;
        for (int i = max(opt_l, m); i <= min(opt_r, n - 1); ++i) if
            (prev[i + 1] != inf) {
            int val = cost(m, i) + prev[i + 1];
            if (val < curr[m]) {
                curr[m] = val;
                opt = i;
            }
        }
        self(self, l, m - 1, opt_l, opt);
        self(self, m + 1, r, opt, opt_r);
    };
    for (int i = 1; i <= n; ++i) {
        curr[i] = cost(i, n);
    }
    for (int i = 2; i <= k; ++i) {
        swap(prev, curr);
        compute(compute, 1, n, 1, n);
    }
    return curr[1];
}
```

2.3 knuth

```
int knuth() {
    vector<vector<int>> val(n + 1, vector<int>(n + 1));
    vector<vector<int>> opt(n + 1, vector<int>(n + 1));
    for (int i = 1; i <= n; ++i) {
        val[i][i] = cost(i, i);
        opt[i][i] = i;
    }
    for (int len = 2; len <= n; ++len) {
        for (int l = 1, r = len; r <= n; ++l, ++r) {
            val[l][r] = inf;
            opt[l][r] = opt[l][r - 1];
            for (int k = opt[l][r - 1]; k <= min(r - 1, opt[l + 1
                ][r]); ++k) {
                int curr = val[l][k] + val[k + 1][r] + cost(l, r);
                if (curr < val[l][r]) {
                    val[l][r] = curr;
                    opt[l][r] = k;
                }
            }
        }
    }
    return val[1][n];
}
```

3.1 divide-and-conquer

```
vector<int> divide_and_conquer() {
    int n, k;
    vector<int> a(n + 1);
    vector<int> pref(n + 2), suf(n + 2);
    vector<int> ans(k + 1);
    auto rec = [&](auto &rec, int low, int high, vector<tuple<int,
        int, int>> &q) -> void {
        if (low > high or q.empty()) return;
        int mid = (low + high) / 2;
        pref[mid + 1] = 0;
        suf[mid] = 0;
        for (int i = mid; i >= low; --i) pref[i] = a[i] + pref[i + 1];
        for (int i = mid + 1; i <= high; ++i) suf[i] = a[i] + suf[i -
            1];
        vector<tuple<int, int, int>> ql, qr;
        for (auto [l, r, i] : q) {
            if (r < mid) ql.push_back({l, r, i});
            else if (l > mid) qr.push_back({l, r, i});
            else ans[i] = pref[l] + suf[r];
        }
        rec(rec, low, mid - 1, ql);
        rec(rec, mid + 1, high, qr);
    };
    return ans;
}
```

3.2 macros

```
void scan(auto&...args) {(cin >> ... >> args);}
void print(auto&&...args) {(cout << ... << args);}
```

3.3 offset-array

```
template <typename T>
struct offset_array {
    int l;
    vector<T> a;

    offset_array(int l, int r, T x = T()): l(l), a(r - l, x) {}

    T &operator[](int i) {
        return a[i - l];
    }

    const T &operator[](int i) const {
        return a[i - l];
    }
};
```

3.4 permutation-int

```
int permutation_to_int(vector<int> &p) {
    int mask = 0;
    int i = 0, r = 0;
    for (int x : p) {
        r = r * ++i + __builtin_popcountll(mask & -(1LL << x));
        mask |= 1LL << x;
    }
    return r;
}
```

3.5 pragmas

```
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

3.6 random

```
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
uniform_int_distribution<int> dist(l, r);
```

3.7 submasks

```
void submasks(int mask) {
    for (int submask = mask; submask > 0; submask = mask & (submask -
        1)) {
        process(submask);
    }
    process(0);
}
```

3.8 sum-over-subsets

```
vector<int> sum_over_subsets(int n, vector<int> f) {
    int full = (1 << n) - 1;
    for (int i = 0; i < n; ++i) {
        int bit = 1 << i;
        for (int mask = 0; mask <= full; ++mask) if (mask & bit) {
            f[mask] += f[mask ^ bit];
        }
    }
    return f;
}
```

3.9 sum-over-supersets

```
vector<int> sum_over_supersets(int n, vector<int> f) {
    int full = (1 << n) - 1;
```

```
    for (int i = 0; i < n; ++i) {
        int bit = 1 << i;
        for (int mask = 0; mask <= full; ++mask) if (not (mask &
             bit)) {
            f[mask] += f[mask | bit];
        }
    }
    return f;
}
```

3.10 ternary-search

```
int ternary_search(int low, int high, bool is_max) {
    while (high - low >= 3) {
        int mid1 = low + (high - low) / 3;
        int mid2 = high - (high - low) / 3;

        if (is_max == (fun(mid1) < fun(mid2))) low = mid1;
        else high = mid2;
    }

    int ans = low;
    for (int i = low + 1; i <= high; ++i) if (is_max == (fun(ans) <
         fun(i))) ans = i;
    return ans;
}

float ternary_search(float low, float high, bool is_max) {
    for (int itr = 1; itr <= 100; ++itr) {
        float mid1 = low + (high - low) / 3.0;
        float mid2 = high - (high - low) / 3.0;

        if (is_max == (fun(mid1) < fun(mid2))) low = mid1;
        else high = mid2;
    }

    return (low + high) / 2.0;
}
```

3.11 time

```
from datetime import date, timedelta

# Convert 'HH:MM:SS' string to total seconds since midnight
def time_to_seconds(time_str: str) -> int:
    h, m, s = map(int, time_str.split(':'))
    return h * 3600 + m * 60 + s

# Convert total seconds back to 'HH:MM:SS' format
def seconds_to_time(seconds: int) -> str:
    h = seconds // 3600
    m = (seconds % 3600) // 60
    s = seconds % 60
    return f"{h:02d}:{m:02d}:{s:02d}"

# Fast parse 'YYYY-MM-DD' directly into integers
def parse_date(date_str: str) -> tuple:
    y, m, d = map(int, date_str.split('-'))
    return y, m, d

# Check if a given date is Saturday or Sunday
def is_weekend(year: int, month: int, day: int) -> bool:
    # weekday() returns 0 for Monday up to 6 for Sunday
    return date(year, month, day).weekday() >= 5

# Get the new (year, month, day) after adding x days
def add_days(year: int, month: int, day: int, x_days: int) -> tuple:
    new_date = date(year, month, day) + timedelta(days=x_days)
    return new_date.year, new_date.month, new_date.day

# Calculate the exact number of days between two dates
def days_between(y1: int, m1: int, d1: int, y2: int, m2: int, d2:
     int) -> int:
    date1 = date(y1, m1, d1)
    date2 = date(y2, m2, d2)
    return (date2 - date1).days

# Fast check if a year is a leap year (pure math, no imports)
def is_leap_year(year: int) -> bool:
    return (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)
```

4.1 circle

```
point incircle(point a, point b, point c) {
    T ab = dist(a, b);
    T bc = dist(b, c);
    T ac = dist(a, c);
    T p = ab + bc + ac;
    if (p == 0) return {};
    return (a * bc + b * ac + c * ab) / p;
}

point circumcircle(point a, point b, point c) {
    point v = b - a;
    point w = c - a;
    T h = cross(v, w);
    if (h == 0) return {};
    return a + perp(v * norm(w) - w * norm(v)) / (2.0 * h);
}

vector<point> circle_line_inter(point o, T r, line l) {
    T s = l.side(o);
    T d2 = (s * s) / norm(l.v);
    T h2 = r * r - d2;
    if (h2 < 0) return {};
    point p = l.proj(o);
    if (h2 == 0) return {p};
    point h = l.v * sqrt(h2 / norm(l.v));
    return {p - h, p + h};
}

vector<point> circle_segment_inter(point o, T r, point a, point b) {
    if (a == b) {
        if (norm(a - o) == r * r) return {a};
        return {};
    }
    vector<point> l = circle_line_inter(o, r, line(a, b));
    vector<point> h;
    for (point p : l) if (in_disk(a, b, p)) h.push_back(p);
    return h;
}

T circle_inter_area(point o1, T r1, point o2, T r2) {
    T d = dist(o1, o2);
    if (d >= r1 + r2) return 0.0;
    if (d <= abs(r1 - r2)) return pi * min(r1, r2) * min(r1, r2);
    T h1 = 2.0 * acos((d * d + r1 * r1 - r2 * r2) / (2.0 * d * r1));
    T h2 = 2.0 * acos((d * d + r2 * r2 - r1 * r1) / (2.0 * d * r2));
    T a1 = 0.5 * r1 * r1 * (h1 - sin(h1));
    T a2 = 0.5 * r2 * r2 * (h2 - sin(h2));
    return a1 + a2;
}

vector<point> circle_inter(point o1, T r1, point o2, T r2) {
    point v = o2 - o1;
    T d2 = norm(v);
    if (d2 == 0) return {};
    T op = (d2 + r1 * r1 - r2 * r2) / 2.0;
    T h2 = r1 * r1 - (op * op) / d2;
    if (h2 < 0) return {};
    point p = o1 + v * (op / d2);
    if (h2 == 0) return {p};
    point h = perp(v) * sqrt(h2 / d2);
    return {p - h, p + h};
}
```

4.2 li-chao-tree

```
// Li Chao Tree: dynamic line & segment container for maximum convex
     hull trick
// Time: line insert/query O(log U) | segment insert O(log^2 U) |
     Space: O(N log U)
// Note: computes max(ax + b) | for min: insert (-a, -b) and take
     -query(x)
struct li_chao_tree {
private:
    struct line {
        int a, b;

        int operator()(int x) const {
            return a * x + b;
        }
    };

    struct node {
        line f;
        int l, r;
    };

    int min_x, max_x;
    vector<node> tree;

    int create_node() {
        tree.push_back({{0, -inf}, 0, 0});
        return tree.size() - 1;
    }

    int insert_line(line f, int p, int l, int r) {
        if (p == 0) p = create_node();
        int m = l + (r - l) / 2;
        bool is_l = f(l) > tree[p].f(l);
        bool is_m = f(m) > tree[p].f(m);
        if (is_m) swap(f, tree[p].f);
        if (l == r) return p;
        if (is_l != is_m) tree[p].l = insert_line(f, tree[p].l, l, m);
        else tree[p].r = insert_line(f, tree[p].r, m + 1, r);
        return p;
    }

    int insert_segment(line f, int ql, int qr, int p, int l, int r) {
        if (p == 0) p = create_node();
        if (ql <= l and r <= qr) return insert_line(f, p, l, r);
        int m = l + (r - l) / 2;
        if (ql <= m) tree[p].l = insert_segment(f, ql, qr, tree[p].l,
             l, m);
        if (qr > m) tree[p].r = insert_segment(f, ql, qr, tree[p].r,
             m + 1, r);
        return p;
    }

    int query(int x, int p, int l, int r) {
        if (p == 0) return -inf;
        int y = tree[p].f(x);
        if (l == r) return y;
        int m = l + (r - l) / 2;
        if (x <= m) y = max(y, query(x, tree[p].l, l, m));
        else y = max(y, query(x, tree[p].r, m + 1, r));
        return y;
    }

public:
    li_chao_tree(int min_x, int max_x): min_x(min_x), max_x(max_x) {
        tree.push_back({{0, -inf}, 0, 0});
        create_node();
    }

    void insert_line(int a, int b) {
        insert_line({a, b}, 1, min_x, max_x);
    }

    void insert_segment(int a, int b, int l, int r) {
```

```
        insert_segment({a, b}, l, r, 1, min_x, max_x);
    }

    int query(int x) {
        return query(x, 1, min_x, max_x);
    }
};
```

4.3 line-container

```
// Line Container: dynamic convex hull trick for maximum
// Time: insert/query O(log N) | Space: O(N)
// Note: computes max(ax + b) | change '<' to '>' in line operator
     for min
struct line {
    mutable int a, b, i;

    bool operator<(const line &o) const {
        return tie(a, b) < tie(o.a, o.b);
    }

    bool operator<(const int x) const {
        return i < x;
    }
};

struct line_container : multiset<line, less<>> {
    const int inf = 1e18;

    int div(int x, int y) {
        return (x / y) - (((x ^ y) < 0) and x % y);
    }

    bool isect(iterator s, iterator t) {
        if (t == end()) return s->i = inf, false;
        if (s->a == t->a) s->i = -inf;
        else s->i = div(t->b - s->b, s->a - t->a);
        return s->i >= t->i;
    }

    void insert(int a, int b) {
        auto z = multiset::insert({a, b, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() and isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() and (--x)->i >= y->i) isect(x,
             erase(y));
    }

    int max(int x) {
        auto l = lower_bound(x);
        return l->a * x + l->b;
    }
};
```

4.4 line

```
struct line {
    point v;
    T c;

    line(point v, T c): v(v), c(c) {}

    line(T a, T b, T c): v({b, -a}), c(c) {}

    line(point p, point q): v(q - p), c(cross(q, p)) {}

    T side(point p) {
        return cross(v, p) - c;
    }

    float dist(point p) {
        return abs(side(p)) / sqrt(norm(v));
    }

    bool comp(point p, point q) {
        return dot(v, p) < dot(v, q);
    }

    // Projection: calculates the orthogonal projection of point p
         onto the line
    // Note: translates p perpendicularly to the line (assumes perp
         is counter-clockwise)
    point proj(point p) {
        return p - (perp(v) * side(p)) / norm(v);
    }

    // Reflection: finds the mirror image of point p across the line
    // Note: projects p and extends it by the same distance
    point refl(point p) {
        return p - (2.0 * perp(v) * side(p)) / norm(v);
    }
};

vector<point> line_inter(line l, line r) {
    if (cross(l.v, r.v) == 0) return {};
    return {(l.c * r.v - r.c * l.v) / cross(l.v, r.v)};
}

vector<point> line_segment_inter(line l, point a, point b) {
    T sa = l.side(a);
    T sb = l.side(b);
    if (sign(sa) * sign(sb) > 0) return {};
    if (sa == 0 and sb == 0) {
        vector<point> p = {a, b};
        sort(p.begin(), p.end());
        p.erase(unique(p.begin(), p.end()), p.end());
        return p;
    }
    return {(a * sb - b * sa) / (sb - sa)};
}
```

4.5 point

```
#define x real()
#define y imag()

using T = int;
using point = complex<T>;

namespace std {
    bool operator<(const point &a, const point &b) {
        return pair(a.x, a.y) < pair(b.x, b.y);
    }
}

istream &operator>>(istream &is, point &p) {
    T r, i; is >> r >> i;
    p = {r, i};
    return is;
}

int sign(T n) {
    if (n == 0) return 0;
    return n < 0 ? -1 : +1;
}

float dist(point a, point b) {
    return sqrt(norm(a - b));
}

T dot(point p, point q) {
    return (conj(p) * q).x;
}

T cross(point p, point q) {
    return (conj(p) * q).y;
}

T orient(point a, point b, point c) {
    return cross(b - a, c - a);
}

point perp(point p) {
    return {-p.y, p.x};
}

// Directed Angle: shortest angle from vector v to vector w
// Note: returns radians in [-pi, pi], negative result means w is to
     the right
float angle(point v, point w) {
    return atan2(cross(v, w), dot(v, w));
}

float closest_pair(vector<point> p) {
    int n = p.size();
    sort(p.begin(), p.end());
    auto compare = [](const point &a, const point &b) {
        if (a.y != b.y) return a.y < b.y;
        return a.x < b.x;
    };
    multiset<point, decltype(compare)> active(compare);
    active.insert(p[0]);
    float d = inf;
    int l = 0;
    for (int i = 1; i < n; ++i) {
        while (l < i and p[i].x - p[l].x >= d) {
            active.erase(active.find(p[l]));
            l++;
        }
        auto begin = active.lower_bound({p[i].x - d, p[i].y - d});
        auto end = active.upper_bound({p[i].x + d, p[i].y + d});
        for (auto itr = begin; itr != end; ++itr) {
            float c = dist(*itr, p[i]);
            d = min(d, c);
        }
        active.insert(p[i]);
    }
    return d;
}
```

4.6 polar-sort

```
// Is Upper: determines which half-plane a point lies in relative to
     ray v
// Note: returns 1 for (0, pi] (upper half and opposite ray), 0 for
     (pi, 2pi] (lower half, ray v and origin)
bool is_upper(point p, point v) {
    T c = cross(v, p);
    T d = dot(v, p);
    return c > 0 or (c == 0 and d < 0);
}

// Polar Sort: sorts points counter-clockwise starting from the
     opposite of reference ray v
// Time: O(N log N) | Space: O(1)
// Note: colinear points sorted by distance
void polar_sort(vector<point> &p, point o = {0, 0}, point v = {1, 0})
     {
    sort(p.begin(), p.end(), [&](point a, point b) {
        a = a - o, b = b - o;
        bool ha = is_upper(a, v), hb = is_upper(b, v);
        if (ha != hb) return ha < hb;
        T c = cross(a, b);
        if (c != 0) return c > 0;
        return norm(a) < norm(b);
    });
}
```

4.7 polygon

```
float area(point a, point b, point c) {
```

```
    return abs(cross(b - a, c - a)) / 2.0;
}

float area(const vector<point> &p) {
    T a = 0;
    int n = p.size();
    for (int i = 0; i < n; ++i) a += cross(p[i], p[(i + 1) % n]);
    return abs(a) / 2.0;
}

// Pick's Theorem: calculates the area of a simple polygon with
    integer vertices
// Time: O(1) | Space: O(1)
// Note: i is the number of interior points, b is the number of
    boundary points
float picks_theorem(int i, int b) {
    return i + (b / 2.0) - 1;
}

// Boundary Points: calculates the total number of integer points on
    the boundary of a polygon
// Time: O(N log M) | Space: O(1)
// Note: relies on the property that the number of integer points on
    a segment is gcd(dx, dy)
int boundary_points(const vector<point> &g) {
    int c = 0, n = g.size();
    for (int i = 0; i < n; ++i) {
        point p = g[i], q = g[(i + 1) % n];
        c += gcd(p.x - q.x, p.y - q.y);
    }
    return c;
}

// Winding Number: computes how many times a polygon wraps around a
    point
// Time: O(N) | Space: O(1)
// Note: returns non-zero for inside, zero for outside or boundary
int winding_number(const vector<point> &g, point a) {
    int w = 0, n = g.size();
    for (int i = 0; i < n; ++i) {
        point p = g[i], q = g[(i + 1) % n];
        if (on_segment(p, q, a)) return 0;
        int d = (q.y >= a.y) - (p.y >= a.y);
        if (d * orient(p, q, a) > 0) w += d;
    }
    return w;
}

vector<point> convex_hull(vector<point> g, bool col) {
    sort(g.begin(), g.end());
    g.erase(unique(g.begin(), g.end()), g.end());
    int n = g.size();
    if (n < 3) return g;
    bool all = col;
    for (int i = 2; i < n and all; ++i) {
        all &= orient(g[0], g[1], g[i]) == 0;
    }
    if (all) return g;
    vector<point> h;
    T lim = col ? 0 : 1;
    for (int i = 0; i <= 1; ++i) {
        int s = h.size();
        for (point p : g) {
            while (h.size() >= s + 2 and orient(h[h.size() - 2],
                 h[h.size() - 1], p) < lim) {
                h.pop_back();
            }
            h.push_back(p);
        }
        h.pop_back();
        reverse(g.begin(), g.end());
    }
    return h;
}
```

4.8 ray

```
float ray_point(point a, point b, point p) {
    if (a != b) {
        line l(a, b);
        if (l.comp(a, p)) return l.dist(p);
    }
    return dist(p, a);
}

float ray_ray(point a, point b, point c, point d) {
    if (a == b) return ray_point(c, d, a);
    if (c == d) return ray_point(a, b, c);
    line l(a, b), r(c, d);
    if (cross(l.v, r.v) != 0) {
        point p = inter(l, r);
        if (l.comp(a, p) and r.comp(c, p)) return 0;
    }
    return min(ray_point(a, b, c), ray_point(c, d, a));
}
```

4.9 segment

```
bool in_disk(point a, point b, point p) {
    return dot(a - p, b - p) <= 0;
}

bool on_segment(point a, point b, point p) {
    return orient(a, b, p) == 0 and in_disk(a, b, p);
}

vector<point> proper_inter(point a, point b, point c, point d) {
    T oa = orient(c, d, a),
      ob = orient(c, d, b),
      oc = orient(a, b, c),
      od = orient(a, b, d);
    if (sign(oa) * sign(ob) >= 0 or sign(oc) * sign(od) >= 0) return
        {};
    return {(a * ob - b * oa) / (ob - oa)};
}

vector<point> inters(point a, point b, point c, point d) {
    vector<point> p = proper_inter(a, b, c, d);
    if (p.size()) return p;
    if (on_segment(c, d, a)) p.push_back(a);
    if (on_segment(c, d, b)) p.push_back(b);
    if (on_segment(a, b, c)) p.push_back(c);
    if (on_segment(a, b, d)) p.push_back(d);
    sort(p.begin(), p.end());
    p.erase(unique(p.begin(), p.end()), p.end());
    return p;
}

float segment_point(point a, point b, point p) {
    if (a != b) {
        line l(a, b);
        if (l.comp(a, p) and l.comp(p, b)) return l.dist(p);
    }
    return min(dist(p, a), dist(p, b));
}

float segment_segment(point a, point b, point c, point d) {
    if (proper_inter(a, b, c, d).size()) return 0;
    return min({segment_point(a, b, c), segment_point(a, b, d),
                segment_point(c, d, a), segment_point(c, d, b)});
}
```

5.1 articulations

```
vector<int> articulations(int one, vector<vector<int>> &g) {
    int n = g.size(), timer = 1;
    vector<int> in(n), low(n);
    vector<bool> art(n);
    auto dfs = [&](auto &self, int a, int p) -> void {
        in[a] = low[a] = timer++;
        int c = 0;
        for (int b : g[a]) if (b != p) {
            if (in[b] != 0) {
                low[a] = min(low[a], in[b]);
            } else {
                c++;
                self(self, b, a);
                low[a] = min(low[a], low[b]);
                if (a != p and low[b] >= in[a]) art[a] = true;
            }
        }
        if (a == p and c > 1) art[a] = true;
    };

    vector<int> points;
    for (int i = one; i < n; ++i) {
        if (in[i] == 0) dfs(dfs, i, i);
        if (art[i]) points.push_back(i);
    }
    return points;
}
```

5.2 block-cut-tree

```
// Block-Cut Tree: decomposes an undirected graph into biconnected
    components
// Time: O(V + E) | Space: O(V + E)
// Note: returns a tree where nodes [0, N) are original vertices and
    nodes >= N are blocks
vector<vector<int>> block_cut_tree(vector<vector<int>> &g) {
    int n = g.size(), t = 0;
    vector<int> tin(n, -1), low(n), st;
    vector<vector<int>> bct(n); bct.reserve(2 * n);
    auto dfs = [&](auto &self, int a, int p) -> void {
        tin[a] = low[a] = t++;
        st.push_back(a);
        bool parent = true;
        for (int b : g[a]) {
            if (p == b and parent) {
                parent = false;
                continue;
            }
            if (tin[b] != -1) {
                low[a] = min(low[a], tin[b]);
            } else {
                self(self, b, a);
                low[a] = min(low[a], low[b]);
                if (tin[a] <= low[b]) {
                    int i = bct.size();
                    bct.emplace_back();
                    bct[i].push_back(a);
                    bct[a].push_back(i);
                    while (true) {
                        int c = st.back(); st.pop_back();
                        bct[i].push_back(c);
                        bct[c].push_back(i);
                        if (b == c) break;
                    }
                }
            }
        }
    };
    for (int i = 0; i < n; ++i) {
        if (tin[i] == -1) {
            dfs(dfs, i, -1);
        }
    }
    return bct;
}
```

5.3 bridge-tree

```
// Bridge Tree: condenses 2-edge-connected components into single
     nodes
// Time: O(V + E) | Space: O(V + E)
// Note: returns a tree and an id array where id[a] is the new node
     id of original vertex a
pair<vector<vector<int>>, vector<int>>
     bridge_tree(vector<vector<int>> &g) {
    int n = g.size(), t = 0;
    vector<int> tin(n, -1), low(n), id(n), st;
    vector<vector<int>> bt;
    auto dfs = [&](auto &self, int a, int p) -> void {
        tin[a] = low[a] = t++;
        st.push_back(a);
        bool parent = true;
        for (int b : g[a]) {
            if (b == p and parent) {
                parent = false;
                continue;
            }
            if (tin[b] != -1) {
                low[a] = min(low[a], tin[b]);
            } else {
                self(self, b, a);
                low[a] = min(low[a], low[b]);
            }
        }
        if (low[a] == tin[a]) {
            int i = bt.size();
            bt.emplace_back();
            while (true) {
                int c = st.back(); st.pop_back();
                id[c] = i;
                if (c == a) break;
            }
        }
    };
    for (int i = 0; i < n; ++i) {
        if (tin[i] == -1) {
            dfs(dfs, i, -1);
        }
    }
    for (int a = 0; a < n; ++a) {
        for (int b : g[a]) {
            if (id[a] != id[b]) {
                bt[id[a]].push_back(id[b]);
            }
        }
    }
    return {bt, id};
}
```

5.4 bridges

```
vector<pair<int, int>> bridges(int one, vector<vector<int>> &g) {
    int n = g.size(), timer = 1;
    vector<int> in(n), low(n);
    vector<pair<int, int>> e;

    auto dfs = [&](auto &self, int a, int p) -> void {
        in[a] = low[a] = timer++;
        for (int b : g[a]) if (b != p) {
            if (in[b] != 0) {
                low[a] = min(low[a], in[b]);
            } else {
                self(self, b, a);
                low[a] = min(low[a], low[b]);
                if (in[a] < low[b]) e.emplace_back(a, b);
            }
        }
    };

    for (int i = one; i < n; ++i) if (in[i] == 0) dfs(dfs, i, i);
    return e;
}
```

5.5 cycle

```
vector<int> find_cycle(bool one, bool is_directed,
     vector<vector<int>> &g) {
    int n = g.size();
    vector<int> state(n), parent(n);
    vector<int> cycle;
    auto dfs = [&](auto &self, int a, int p) -> bool {
        state[a] = 1;
        for (int b : g[a]) if (b != p or is_directed) {
            if (state[b] == 0) {
                parent[b] = a;
                if (self(self, b, a)) return true;
            } else if (state[b] == 1) {
                int c = a;
                while (c != b) {
                    cycle.push_back(c);
                    c = parent[c];
                }
                cycle.push_back(b);
                reverse(cycle.begin(), cycle.end());
                return true;
            }
        }
        state[a] = 2;
        return false;
    };
    for (int i = one; i < n; ++i) if (state[i] == 0) {
        if (dfs(dfs, i, -1)) return cycle;
    }
    return {};
}
```

5.6 dominator-tree

```
// Dominator Tree: computes immediate dominators in directed graphs
// Time: build O(V log V + E) | Space: O(V + E)
// Note: returns the tree where node a immediately dominates its
     children
struct dominator_dsu {
    vector<int> p, c;
    vector<int> &x;

    dominator_dsu(int n, vector<int> &x): p(n), c(n), x(x) {
        iota(p.begin(), p.end(), 0);
        iota(c.begin(), c.end(), 0);
    }

    int find(int a) {
        if (p[a] == a) return c[a];
        int r = p[a];
        if (p[r] != r) {
            find(r);
            if (x[c[r]] < x[c[a]]) c[a] = c[r];
            p[a] = p[r];
        }
        return c[a];
    }

    void unite(int a, int b) {
        p[b] = a;
    }
};

vector<vector<int>> dominator_tree(int s, vector<vector<int>> &g) {
    int n = g.size();
    vector<int> tin(n, -1), inv(n), parent(n);
    int t = 0;
    auto dfs = [&](auto &self, int a) -> void {
        tin[a] = t;
        inv[t] = a;
        t++;
        for (int b : g[a]) if (tin[b] == -1) {
            parent[b] = a;
            self(self, b);
        }
    };
    dfs(dfs, s);
    vector<vector<int>> rev(n);
    for (int a = 0; a < n; ++a) {
        for (int b : g[a]) rev[b].push_back(a);
    }
    vector<int> semi(n), dom(n);
    for (int a = 0; a < n; ++a) semi[a] = tin[a];
    vector<vector<int>> bucket(n);
    dominator_dsu dsu(n, semi);
    for (int i = t - 1; i >= 1; --i) {
        int a = inv[i];
        for (int b : rev[a]) if (tin[b] != -1) {
            semi[a] = min(semi[a], semi[dsu.find(b)]);
        }
        bucket[inv[semi[a]]].push_back(a);
        int p = parent[a];
        dsu.unite(p, a);
        for (int b : bucket[p]) {
            dom[b] = p;
            int c = dsu.find(b);
            if (semi[c] < semi[b]) dom[b] = c;
        }
        bucket[p].clear();
    }
    vector<vector<int>> dt(n);
    for (int i = 1; i < t; ++i) {
        int a = inv[i];
        if (dom[a] != inv[semi[a]]) dom[a] = dom[dom[a]];
        dt[dom[a]].push_back(a);
    }
    return dt;
}
```

5.7 steiner-tree

```
vector<vector<int>> steiner_tree(int n, int k, const
     vector<vector<int>> &dist) {
    vector<vector<int>> steiner(1 << k, vector<int>(n, inf));
    for (int i = 0; i < n; ++i) steiner[0][i] = 0;
    for (int i = 0; i < k; ++i) steiner[1 << i][i] = 0;
    for (int mask = 1; mask < (1 << k); ++mask) {
        for (int sub = mask; sub > 0; sub = mask & (sub - 1)) {
            for (int i = 0; i < n; ++i) {
                steiner[mask][i] = min(steiner[mask][i],
                     steiner[sub][i] + steiner[mask ^ sub][i]);
            }
        }
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < n; ++j) {
                steiner[mask][j] = min(steiner[mask][j],
                     steiner[mask][i] + dist[i][j]);
            }
        }
    }
    return steiner;
}
```

5.8 tarjan

```
vector<vector<int>> tarjan(int r, vector<vector<int>> &g) {
    int n = g.size(), timer = 1;
    vector<vector<int>> components;
    vector<int> id(n), low(n);
    stack<int> trail;
    vector<bool> in(n);

    auto dfs = [&](auto &self, int a) -> void {
        id[a] = low[a] = timer++;
```

```
        trail.push(a);
        in[a] = true;
        for (int b : g[a]) {
            if (id[b] == 0) {
                self(self, b);
                low[a] = min(low[a], low[b]);
            } else if (in[b]) {
                low[a] = min(low[a], id[b]);
            }
        }

        if (id[a] == low[a]) {
            vector<int> component;
            while (true) {
                int b = trail.top(); trail.pop();
                in[b] = false;
                component.push_back(b);
                if (b == a) break;
            }
            components.push_back(component);
        }
    };

    for (int a = r; a < n; ++a) {
        if (id[a] == 0) {
            dfs(dfs, a);
        }
    }

    reverse(components.begin(), components.end());
    return components;
}

pair<vector<vector<int>>, vector<int>> condensed_graph(int r,
     vector<vector<int>> &g) {
    int n = g.size();
    vector<vector<int>> components = tarjan(r, g);
    int m = components.size();

    vector<int> id(n);
    for (int i = 0; i < m; ++i) {
        for (int a : components[i]) {
            id[a] = i;
        }
    }

    vector<vector<int>> c(m);
    for (int a = r; a < n; ++a) {
        for (int b : g[a]) if (id[a] != id[b]) {
            c[id[a]].push_back(id[b]);
        }
    }

    for (int i = 0; i < m; ++i) {
        sort(c[i].begin(), c[i].end());
        c[i].erase(unique(c[i].begin(), c[i].end()), c[i].end());
    }
    return {c, id};
}
```

5.9 bellman-ford

```
vector<int> bellman_ford(int s, int n, vector<array<int, 3>> &e) {
    vector<int> d(n, inf);
    d[s] = 0;
    for (int i = 1; i < n; ++i) {
        bool chd = false;
        for (auto [a, b, w] : e) if (d[a] != inf and d[a] + w < d[b])
             {
            d[b] = d[a] + w;
            chd = true;
        }
        if (not chd) break;
    }
    for (auto [a, b, w] : e) if (d[a] != inf and d[a] + w < d[b])
         return {};
    return d;
}

vector<int> spfa(int s, vector<vector<pair<int, int>>> &g) {
    int n = g.size();
    vector<int> d(n, inf), c(n);
    vector<bool> in(n);
    queue<int> q;
    d[s] = 0;
    in[s] = true;
    q.push(s);
    while (q.size()) {
        int a = q.front(); q.pop();
        in[a] = false;
        for (auto [b, w] : g[a]) if (d[a] + w < d[b]) {
            d[b] = d[a] + w;
            if (not in[b]) {
                c[b] += 1;
                if (c[b] >= n) return {};
                in[b] = true;
                q.push(b);
            }
        }
    }
    return d;
}
```

5.10 floyd-warshall

```
vector<vector<int>> floyd_warshall(int n, vector<vector<int>> &g) {
    vector<vector<int>> d = g;
    for (int i = 0; i < n; ++i) {
        d[i][i] = 0;
    }
    for (int k = 0; k < n; ++k) {
        for (int i = 0; i < n; ++i) if (d[i][k] != inf) {
            for (int j = 0; j < n; ++j) if (d[k][j] != inf) {
                d[i][j] = min(d[i][j], d[i][k] + d[k][j]);
            }
        }
    }
    return d;
}
```

5.11 johnson

```
vector<vector<int>> johnson(vector<vector<pair<int, int>>> &g) {
    int n = g.size();
    g.emplace_back();
    for (int i = 0; i < n; ++i) {
        g[n].emplace_back(i, 0);
    }
    vector<int> h = spfa(n, g);
    g.pop_back();
    if (h.empty()) return {};
    h.pop_back();
    for (int i = 0; i < n; ++i) {
        for (auto &[j, w] : g[i]) {
            w += h[i] - h[j];
        }
    }
    vector<vector<int>> d;
    for (int i = 0; i < n; ++i) {
        d.push_back(dijkstra(i, g));
        for (int j = 0; j < n; ++j) {
            if (d[i][j] != inf) d[i][j] -= h[i] - h[j];
        }
    }
    return d;
}
```

5.12 dinic

```
struct dinic {
    struct edge {
        int b, w, i;
        bool h;
    };

    int n;
    vector<vector<edge>> g;
    vector<int> l, p;

    dinic(int n): n(n), g(n), l(n), p(n) {}

    void insert_directed(int a, int b, int w) {
        int i = g[b].size();
        int j = g[a].size();
        g[a].push_back({b, w, i, true});
        g[b].push_back({a, 0, j, false});
    }

    void insert_undirected(int a, int b, int w) {
        int i = g[b].size();
        int j = g[a].size();
        g[a].push_back({b, w, i, true});
        g[b].push_back({a, w, j, true});
    }

    bool bfs(int s, int t) {
        fill(l.begin(), l.end(), -1);
        l[s] = 0;
        queue<int> q;
        q.push(s);

        while (not q.empty()) {
            int a = q.front(); q.pop();
            for (auto &[b, w, i, h] : g[a]) {
                if (w > 0 and l[b] < 0) {
                    l[b] = l[a] + 1;
                    q.push(b);
                }
            }
        }
        return l[t] >= 0;
    }

    int dfs(int a, int t, int f) {
        if (a == t or f == 0) return f;

        for (int &k = p[a]; k < g[a].size(); ++k) {
            auto &[b, w, i, h] = g[a][k];
            if (w > 0 and l[b] == l[a] + 1) {
                if (int d = dfs(b, t, min(f, w))) {
                    w -= d;
                    g[b][i].w += d;
                    return d;
                }
            }
        }
        return 0;
    }

    int flow(int s, int t) {
        int f = 0;
        while (bfs(s, t)) {
            fill(p.begin(), p.end(), 0);
            while (int d = dfs(s, t, inf)) {
                f += d;
            }
        }
        return f;
    }
```

```
    bool on_cut(int a) {
        return l[a] >= 0;
    }

    vector<pair<int, int>> saturated_edges() {
        vector<pair<int, int>> e;
        for (int a = 0; a < n; ++a) {
            for (auto [b, w, i, h] : g[a]) if (h and w == 0) {
                e.push_back({a, b});
            }
        }
        return e;
    }
};
```

5.13 min-cost-k-flow

```
// Min Cost K-Flow: successive shortest path with potentials
// Time: O(F * E log V) where F is flow | Space: O(V + E)
// Note: // Note: SPFA handles initial negative edge costs. If all
     costs are non-negative, replace spfa(s) with fill(h.begin(),
     h.end(), 0)
struct min_cost_k_flow {
    struct edge {
        int b, i, w, c;
        bool o;
    };

    int n;
    vector<vector<edge>> g;
    vector<int> d, h, p, e;

    min_cost_k_flow(int n): n(n), g(n), d(n), h(n), p(n), e(n) {}

    void insert_edge(int a, int b, int w, int c) {
        int i = g[b].size();
        int j = g[a].size();
        g[a].push_back({b, i, w, c, true});
        g[b].push_back({a, j, 0, -c, false});
    }

    bool spfa(int s) {
        fill(h.begin(), h.end(), inf);
        vector<int> cnt(n);
        vector<bool> in(n);
        queue<int> q;
        h[s] = 0;
        in[s] = true;
        q.push(s);
        while (q.size()) {
            int a = q.front(); q.pop();
            in[a] = false;
            for (auto &[b, i, w, c, o] : g[a]) if (w > 0 and h[a] + c
                 < h[b]) {
                h[b] = h[a] + c;
                if (not in[b]) {
                    cnt[b] += 1;
                    if (cnt[b] >= n) return false;
                    in[b] = true;
                    q.push(b);
                }
            }
        }
        return true;
    }

    bool dijkstra(int s, int t) {
        fill(d.begin(), d.end(), inf);
        priority_queue<pair<int, int>, vector<pair<int, int>>,
             greater<pair<int, int>>> pq;
        d[s] = 0;
        pq.emplace(0, s);
        while (pq.size()) {
            auto [x, a] = pq.top(); pq.pop();
            if (d[a] < x) continue;

            for (int i = 0; i < g[a].size(); ++i) {
                auto &[b, j, w, c, o] = g[a][i];
                if (w <= 0) continue;

                int v = d[a] + c + h[a] - h[b];
                if (v < d[b]) {
                    d[b] = v;
                    p[b] = a;
                    e[b] = i;
                    pq.emplace(d[b], b);
                }
            }
        }
        return d[t] != inf;
    }

    pair<int, int> cost_flow(int s, int t, int k = inf) {
        int flow = 0, cost = 0;
        if (not spfa(s)) return {-1, -1};
        while (flow < k and dijkstra(s, t)) {
            for (int a = 0; a < n; ++a) if (d[a] != inf) {
                h[a] += d[a];
            }
            int push = k - flow;
            for (int a = t; a != s; a = p[a]) {
                int b = p[a], i = e[a];
                push = min(push, g[b][i].w);
            }
            for (int a = t; a != s; a = p[a]) {
                int b = p[a], i = e[a], j = g[b][i].i;
                g[b][i].w -= push;
                g[a][j].w += push;
                cost += g[b][i].c * push;
            }
            flow += push;
        }
        return {flow, cost};
    }

    vector<pair<int, int>> saturated_edges() {
        vector<pair<int, int>> edges;
        for (int a = 0; a < n; ++a) {
            for (auto &[b, i, w, c, o] : g[a]) if (w == 0 and o) {
                edges.push_back({a, b});
            }
        }
        return edges;
    }
};
```

5.14 hierholzer

```
// Undirected Eulerian Path: Hierholzer's algorithm for paths and
     circuits
// Time: O(V + E) | Space: O(V + E)
// Note: returns sequence of vertices or empty if impossible | s must
     be a valid start node
vector<int> undirected_eulerian_path(int s, vector<vector<pair<int,
     int>>> &g, int e) {
    int n = g.size();
    int odd = 0;
    for (int i = 0; i < n; ++i) {
        if (g[i].size() & 1) odd++;
    }
    if (odd != 0 and odd != 2) return {};
    if (odd == 2 and (g[s].size() & 1) == 0) return {};
    vector<bool> used(e + 1);
    vector<int> next(n), path;
    auto hierholzer = [&](auto &self, int a) -> void {
        while (next[a] < g[a].size()) {
            auto [b, id] = g[a][next[a]];
            next[a]++;
            if (not used[id]) {
                used[id] = true;
                self(self, b);
            }
        }
        path.push_back(a);
    };
    hierholzer(hierholzer, s);
    if (path.size() != e + 1) return {};
    reverse(path.begin(), path.end());
    return path;
}

// Directed Eulerian Path: Hierholzer's algorithm for paths and
     circuits
// Time: O(V + E) | Space: O(V + E)
// Note: returns sequence of vertices or empty if impossible | s must
     be a valid start node
vector<int> directed_eulerian_path(int s, vector<vector<int>> &g, int
     e) {
    int n = g.size();
    vector<int> in(n), out(n);
    for (int a = 0; a < n; ++a) {
        out[a] = g[a].size();
        for (int b : g[a]) {
            in[b]++;
        }
    }
    int start = 0, end = 0;
    for (int i = 0; i < n; ++i) {
        if (out[i] - in[i] == +1) start++;
        else if (out[i] - in[i] == -1) end++;
        else if (out[i] - in[i] != 0) return {};
    }
    if (not ((start == 0 and end == 0) or (start == 1 and end == 1)))
         return {};
    if (start == 1 and out[s] - in[s] != +1) return {};
    vector<int> next(n), path;
    auto hierholzer = [&](auto &self, int a) -> void {
        while (next[a] < g[a].size()) {
            int b = g[a][next[a]];
            next[a]++;
            self(self, b);
        }
        path.push_back(a);
    };
    hierholzer(hierholzer, s);
    if (path.size() != e + 1) return {};
    reverse(path.begin(), path.end());
    return path;
}
```

5.15 link-cut-tree

```
struct link_cut_tree {
    splay_tree st;

    void init(vector<int> &v) {
        st.init();
        for (int i = 1; i < v.size(); ++i) {
            st.t.emplace_back();
            st.t.back().init(v[i]);
        }
    }

    int access(int a) {
        int p = 0;
        for (int c = a; c != 0; c = st.t[c].p) {
            st.splay(c, 0);
            st.t[c].c[1] = p;
            st.pull(c);
```

```
            p = c;
        }
        st.splay(a, 0);
        return p;
    }

    void make_root(int a) {
        access(a);
        st.flip(a);
    }

    int find_root(int a) {
        access(a);
        while (st.t[a].c[0] != 0) {
            st.push(a);
            a = st.t[a].c[0];
        }
        st.splay(a, 0);
        return a;
    }

    void link(int a, int b) {
        make_root(a);
        if (find_root(b) != a) {
            st.t[a].p = b;
        }
    }

    void cut(int a, int b) {
        make_root(a);
        access(b);
        if (st.t[b].c[0] == a and st.t[a].c[1] == 0) {
            st.t[b].c[0] = 0;
            st.t[a].p = 0;
            st.pull(b);
        }
    }

    int lca(int a, int b) {
        if (find_root(a) != find_root(b)) return 0;
        access(a);
        return access(b);
    }

    void update(int a, int v) {
        st.splay(a, 0);
        st.t[a].val += v;
        st.pull(a);
    }

    int query(int a, int b) {
        make_root(a);
        access(b);
        return st.t[b].sum;
    }
};
```

5.16 hopcroft-karp

```
struct hopcroft_karp {
    const int inf = 1e9;
    int n, m, s;
    vector<int> r, l, d;
    vector<vector<int>> g;

    hopcroft_karp(int n, int m): n(n), m(m), r(n, -1), l(m, -1),
         d(n), g(n) {}

    void edge(int a, int b) {
        g[a].push_back(b);
    }

    bool bfs() {
        s = inf;
        queue<int> q;
        for (int a = 0; a < n; ++a) {
            if (r[a] == -1) {
                d[a] = 0;
                q.push(a);
            } else {
                d[a] = inf;
            }
        }
        while (q.size()) {
            int a = q.front(); q.pop();
            if (d[a] >= s) continue;
            for (int b : g[a]) {
                if (l[b] == -1) {
                    s = min(s, d[a] + 1);
                } else if (d[l[b]] == inf) {
                    d[l[b]] = d[a] + 1;
                    q.push(l[b]);
                }
            }
        }
        return s != inf;
    }

    bool dfs(int a) {
        if (a == -1) return true;
        for (int b : g[a]) {
            int h = (l[b] != -1) ? d[l[b]] : s;
            if (h == d[a] + 1) if (dfs(l[b])) {
                r[a] = b;
                l[b] = a;
                return true;
            }
        }
        d[a] = inf;
        return false;
    }

    int match() {
        int matching = 0;
        while (bfs()) {
            for (int a = 0; a < n; ++a) if (r[a] == -1 and dfs(a))
                 matching++;
        }
        return matching;
    }

    vector<pair<int, int>> matches() {
        vector<pair<int, int>> m;
        for (int a = 0; a < n; ++a) if (r[a] != -1) m.emplace_back(a,
             r[a]);
        return m;
    }
};
```

5.17 hungarian

```
// Hungarian Algorithm: minimum weight bipartite matching
// Time: O(N^2 * M) | Space: O(N + M)
// Note: computes min cost assignment | negate input matrix for max
pair<int, vector<int>> hungarian(vector<vector<int>> g) {
    int n = g.size(), m = g[0].size();
    bool transposed = false;
    if (n > m) {
        transposed = true;
        vector<vector<int>> t(m, vector<int>(n));
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < m; ++j) {
                t[j][i] = g[i][j];
            }
        }
        swap(n, m);
        swap(g, t);
    }

    vector<int> pot_row(n + 1), pot_col(m + 1);
    vector<int> match(m + 1), prev_col(m + 1);

    for (int i = 1; i <= n; ++i) {
        match[0] = i;
        int col = 0;
        vector<int> min_cost(m + 1, inf);
        vector<bool> v(m + 1);
        do {
            v[col] = true;
            int row = match[col];
            int delta = inf;
            int next_col = 0;
            for (int j = 1; j <= m; ++j) if (not v[j]) {
                int cost = g[row - 1][j - 1] - pot_row[row] -
                     pot_col[j];
                if (cost < min_cost[j]) {
                    min_cost[j] = cost;
                    prev_col[j] = col;
                }
                if (min_cost[j] < delta) {
                    delta = min_cost[j];
                    next_col = j;
                }
            }
            for (int j = 0; j <= m; ++j) {
                if (v[j]) {
                    pot_row[match[j]] += delta;
                    pot_col[j] -= delta;
                } else {
                    min_cost[j] -= delta;
                }
            }
            col = next_col;
        } while (match[col] != 0);
        do {
            int prev = prev_col[col];
            match[col] = match[prev];
            col = prev;
        } while (col != 0);
    }

    int cost = -pot_col[0];
    vector<int> answer(n, -1);
    for (int j = 1; j <= m; ++j) if (match[j] != 0) {
        answer[match[j] - 1] = j - 1;
    }

    if (transposed) {
        vector<int> real(m, -1);
        for (int i = 0; i < n; ++i) if (answer[i] != -1) {
            real[answer[i]] = i;
        }
        swap(answer, real);
    }
    return {cost, answer};
}
```

5.18 kuhn

```
// Kuhn's Algorithm: maximum bipartite matching
// Time: O(V * E) | Space: O(N + M)
// Note: connects left [0..n] to right [0..m] | match[right] returns
     left node or -1
vector<int> kuhn(int n, int m, vector<vector<int>> &g) {
    vector<int> match(m + 1, -1);
    int itr = 1;
    vector<int> v(n + 1);

    auto dfs = [&](auto &self, int a) -> bool {
```

```
        if (v[a] == itr) return false;
        v[a] = itr;
        for (int b : g[a]) if (match[b] == -1 or self(self,
             match[b])) {
            match[b] = a;
            return true;
        }
        return false;
    };

    for (int a = 0; a <= n; ++a, ++itr) {
        dfs(dfs, a);
    }
    return match;
}
```

5.19 maximum-clique

```
vector<int> maximum_clique(vector<vector<int>> &g) {
    int n = g.size();
    vector<int> h(n);
    for (int a = 0; a < n; ++a) {
        for (int b : g[a]) {
            h[a] |= 1LL << b;
        }
    }
    int size = 0, mask = 0;
    auto bron_kerbosch = [&](auto &self, int r, int p, int x) -> void
         {
        if (p == 0 and x == 0) {
            int s = __builtin_popcountll(r);
            if (s > size) {
                size = s;
                mask = r;
            }
            return;
        }
        int i = __builtin_ctzll(p | x);
        int c = p & ~h[i];
        while (c) {
            int k = __builtin_ctzll(c);
            int k_mask = 1LL << k;
            self(self, r | k_mask, p & h[k], x & h[k]);
            p ^= k_mask;
            x |= k_mask;
            c ^= k_mask;
        }
    };
    int initial = (1LL << n) - 1;
    bron_kerbosch(bron_kerbosch, 0, initial, 0);
    vector<int> clique;
    for (int i = 0; i < n; ++i) {
        if ((mask >> i) & 1) {
            clique.push_back(i);
        }
    }
    return clique;
}
```

5.20 binary-lifting

```
struct binary_lifting {
    int n, log;
    vector<int> d;
    vector<vector<int>> up;

    void build(int a, int p, const vector<vector<int>> &g) {
        up[a][0] = p;
        for (int i = 1; i <= log; ++i) {
            up[a][i] = up[up[a][i - 1]][i - 1];
        }
        for (int b : g[a]) if (b != p) {
            d[b] = d[a] + 1;
            build(b, a, g);
        }
    }

    void init(const vector<vector<int>> &g) {
        n = g.size();
        log = __bit_width(n) - 1;
        d.assign(n, -1);
        up.assign(n, vector<int>(log + 1));
        for (int i = 0; i < n; ++i) if (d[i] == -1) {
            d[i] = 0;
            build(i, i, g);
        }
    }

    int kth(int a, int k) const {
        k = min(k, d[a]);
        while (k > 0) {
            int i = __bit_width(k) - 1;
            a = up[a][i];
            k ^= 1 << i;
        }
        return a;
    }

    int lca(int a, int b) const {
        if (d[a] < d[b]) swap(a, b);
        a = kth(a, d[a] - d[b]);
        if (a == b) return a;
        for (int i = log; i >= 0; --i) {
            if (up[a][i] != up[b][i]) {
                a = up[a][i];
                b = up[b][i];
            }
        }
        return up[a][0];
    }

    int dist(int a, int b) const {
        return d[a] + d[b] - 2 * d[lca(a, b)];
    }

    int path(int a, int b, int k) const {
        int p = lca(a, b);
        int l = d[a] - d[p];
        int r = d[b] - d[p];
        k = min(k, l + r);
        if (k <= l) return kth(a, k);
        else return kth(b, l + r - k);
    }
};
```

5.21 centroid-decomposition

```
pair<vector<int>, vector<vector<int>>> centroid_decomposition(int r,
     vector<vector<int>> &g) {
    int n = g.size();
    vector<int> parent(n);
    vector<int> size(n);
    vector<bool> removed(n);
    vector<vector<int>> dist(n);
    auto dfs = [&](auto &self, int a, int p) -> int {
        size[a] = 1;
        for (int b : g[a]) if (b != p and not removed[b]) {
            size[a] += self(self, b, a);
        }
        return size[a];
    };
    auto centroid = [&](auto &self, int a, int p, int s) -> int {
        for (int b : g[a]) if ((b != p and not removed[b]) and
             size[b] > s / 2) {
            return self(self, b, a, s);
        }
        return a;
    };
    auto distance = [&](auto &self, int a, int p, int d) -> void {
        dist[a].push_back(d);
        for (int b : g[a]) if (b != p and not removed[b]) {
            self(self, b, a, d + 1);
        }
    };
    auto build = [&](auto &self, int a, int p) -> void {
        int s = dfs(dfs, a, a);
        a = centroid(centroid, a, a, s);
        distance(distance, a, a, 0);
        parent[a] = p;
        removed[a] = true;
        for (int b : g[a]) if (not removed[b]) {
            self(self, b, a);
        }
    };
    build(build, r, -1);
    for (auto &path : dist) reverse(path.begin(), path.end());
    return {parent, dist};
}
```

5.22 heavy-light-decomposition

```
struct hld {
    int n;
    bool e;
    segment_tree s;
    vector<int> p, d, h, id;

    hld(vector<vector<int>> &g, bool e): n(g.size()), e(e), s(n) {
        p.assign(n, -1);
        d.assign(n, 0);
        h.assign(n, 0);
        id.assign(n, 0);
        int t = 0;
        for (int i = 0; i < n; ++i) {
            if (p[i] == -1) {
                p[i] = i;
                d[i] = 0;
                dfs(i, g);
                h[i] = i;
                decompose(i, t, g);
            }
        }
    }

    int dfs(int a, vector<vector<int>> &g) {
        int size = 1, max_sub = 0;
        for (int &b : g[a]) if (b != p[a]) {
            p[b] = a;
            d[b] = d[a] + 1;
            int sub = dfs(b, g);
            if (sub > max_sub) {
                max_sub = sub;
                swap(g[a][0], b);
            }
            size += sub;
        }
        return size;
    }

    void decompose(int a, int &t, const vector<vector<int>> &g) {
        id[a] = t++;
        for (int b : g[a]) if (b != p[a]) {
            h[b] = b;
            if (b == g[a][0]) h[b] = h[a];
            decompose(b, t, g);
        }
    }

    int lca(int a, int b) const {
        while (h[a] != h[b]) {
```

```
            if (d[h[a]] < d[h[b]]) swap(a, b);
            a = p[h[a]];
        }
        if (d[a] > d[b]) swap(a, b);
        return a;
    }

    int query(int a, int b) const {
        int r = s.neutral;
        while (h[a] != h[b]) {
            if (d[h[a]] < d[h[b]]) swap(a, b);
            r = s.merge(r, s.query(id[h[a]], id[a]));
            a = p[h[a]];
        }
        if (d[a] > d[b]) swap(a, b);
        if (e and a == b) return r;
        r = s.merge(r, s.query(id[a] + e, id[b]));
        return r;
    }

    void update(int a, int x) {
        s.update(id[a], x);
    }
};
```

5.23 lazy-heavy-light-decomposition

```
struct lazy_hld {
    int n;
    bool e;
    lazy_segment_tree s;
    vector<int> p, d, h, id;

    lazy_hld(vector<vector<int>> &g, bool e): n(g.size()), e(e), s(n)
        {
        p.assign(n, -1);
        d.assign(n, 0);
        h.assign(n, 0);
        id.assign(n, 0);
        int t = 0;
        for (int i = 0; i < n; ++i) {
            if (p[i] == -1) {
                p[i] = i;
                d[i] = 0;
                dfs(i, g);
                h[i] = i;
                decompose(i, t, g);
            }
        }
    }

    int dfs(int a, vector<vector<int>> &g) {
        int size = 1, max = 0;
        for (int &b : g[a]) if (b != p[a]) {
            p[b] = a;
            d[b] = d[a] + 1;
            int curr = dfs(b, g);
            if (curr > max) {
                max = curr;
                swap(g[a][0], b);
            }
            size += curr;
        }
        return size;
    }

    void decompose(int a, int &t, vector<vector<int>> &g) {
        id[a] = t++;
        for (int b : g[a]) if (b != p[a]) {
            h[b] = b;
            if (b == g[a][0]) h[b] = h[a];
            decompose(b, t, g);
        }
    }

    int lca(int a, int b) {
        while (h[a] != h[b]) {
            if (d[h[a]] < d[h[b]]) swap(a, b);
            a = p[h[a]];
        }
        if (d[a] > d[b]) swap(a, b);
        return a;
    }

    int query(int a, int b) {
        int r = s.neutral;
        while (h[a] != h[b]) {
            if (d[h[a]] < d[h[b]]) swap(a, b);
            r = s.merge(r, s.query(id[h[a]], id[a]));
            a = p[h[a]];
        }
        if (d[a] > d[b]) swap(a, b);
        if (e and a == b) return r;
        r = s.merge(r, s.query(id[a] + e, id[b]));
        return r;
    }

    void update(int a, int b, int x) {
        while (h[a] != h[b]) {
            if (d[h[a]] < d[h[b]]) swap(a, b);
            s.update(id[h[a]], id[a], x);
            a = p[h[a]];
        }
        if (d[a] > d[b]) swap(a, b);
        if (e and a == b) return;
        s.update(id[a] + e, id[b], x);
    }
};
```

5.24 leaf-peeling

```
// Leaf Peeling: finds the center(s) and diameter of an undirected
    tree
// Time: O(N) | Space: O(N)
// Note: returns {diameter, centers} | 'i' sets 0-based or 1-based
    indexing
pair<int, vector<int>> leaf_peeling(int i, vector<vector<int>> &g) {
    int n = g.size() - i;
    if (n == 1) return {0, {i}};
    vector<int> d(g.size());
    queue<int> q;
    for (int a = i; a < g.size(); ++a) {
        d[a] = g[a].size();
        if (d[a] == 1) q.push(a);
    }
    int r = n, l = 0;
    while (r > 2) {
        int s = q.size();
        r -= s;
        l++;
        while (s--) {
            int a = q.front(); q.pop();
            for (int b : g[a]) {
                if (--d[b] == 1) q.push(b);
            }
        }
    }
    vector<int> c;
    while (q.size()) c.push_back(q.front()), q.pop();
    return {2 * l + (c.size() == 2), c};
}
```

5.25 reversible-heavy-light-decomposition

```
struct reversible_segment_tree {
    int n;
    bool v;
    vector<int> tree;

    int merge(int x, int y) {
        return x + y;
    }

    int combine(int x, int y) {
        if (v) return merge(y, x);
        return merge(x, y);
    }

    reversible_segment_tree(int s, bool v): n(__bit_ceil(s)), v(v),
        tree(2 * n) {}

    reversible_segment_tree(vector<int> &a, bool v):
        reversible_segment_tree(a.size(), v) {
        for (int i = 0; i < a.size(); ++i) tree[n + i] = a[i];
        for (int i = n - 1; i > 0; --i) tree[i] = combine(tree[i << 1
            ], tree[(i << 1) | 1]);
    }

    void update(int i, int x) {
        tree[i += n] = x;
        while (i >>= 1) tree[i] = combine(tree[i << 1], tree[(i << 1)
            | 1]);
    }

    int query(int i, int j) {
        int l = 0, r = 0;
        for (i += n, j += n + 1; i < j; i >>= 1, j >>= 1) {
            if (i & 1) l = combine(l, tree[i++]);
            if (j & 1) r = combine(tree[--j], r);
        }
        return combine(l, r);
    }
};

struct reversible_hld {
    int n;
    bool e;
    reversible_segment_tree s, t;
    vector<int> p, d, h, id;

    reversible_hld(vector<vector<int>> &g, bool e): n(g.size()),
        e(e), s(n, false), t(n, true) {
        p.assign(n, -1);
        d.assign(n, 0);
        h.assign(n, 0);
        id.assign(n, 0);
        int k = 0;
        for (int i = 0; i < n; ++i) {
            if (p[i] == -1) {
                p[i] = i;
                d[i] = 0;
                dfs(i, g);
                h[i] = i;
                decompose(i, k, g);
            }
        }
    }

    int dfs(int a, vector<vector<int>> &g) {
        int size = 1, max = 0;
        for (int &b : g[a]) if (b != p[a]) {
            p[b] = a;
            d[b] = d[a] + 1;
            int curr = dfs(b, g);
            if (curr > max) {
                max = curr;
                swap(g[a][0], b);
            }
            size += curr;
        }
```

```
        return size;
    }

    void decompose(int a, int &k, vector<vector<int>> &g) {
        id[a] = k++;
        for (int b : g[a]) if (b != p[a]) {
            h[b] = b;
            if (b == g[a][0]) h[b] = h[a];
            decompose(b, k, g);
        }
    }

    int lca(int a, int b) {
        while (h[a] != h[b]) {
            if (d[h[a]] < d[h[b]]) swap(a, b);
            a = p[h[a]];
        }
        if (d[a] > d[b]) swap(a, b);
        return a;
    }

    int query(int a, int b) {
        int l = s.neutral;
        int r = s.neutral;
        while (h[a] != h[b]) {
            if (d[h[a]] > d[h[b]]) {
                l = t.merge(l, t.query(id[h[a]], id[a]));
                a = p[h[a]];
            } else {
                r = s.merge(s.query(id[h[b]], id[b]), r);
                b = p[h[b]];
            }
        }
        if (e and a == b) return s.merge(l, r);
        if (d[a] > d[b]) {
            l = t.merge(l, t.query(id[b] + e, id[a]));
        } else {
            r = s.merge(s.query(id[a] + e, id[b]), r);
        }
        return s.merge(l, r);
    }

    void update(int a, int x) {
        s.update(id[a], x);
        t.update(id[a], x);
    }
};
```

5.26 two-sat

```
// 2-SAT: Solves Boolean Satisfiability for 2-CNF Formulas
// Time: Build O(N + M) | Space: O(N + M)
// Note: Uses Tarjan's SCC | 0-indexed variables
struct two_sat {
    int n;
    vector<vector<int>> g;

    two_sat(int n): n(n), g(2 * n) {}

    // A or B
    void disjunction(int a, bool n_a, int b, bool n_b) {
        a = (2 * a) ^ n_a;
        b = (2 * b) ^ n_b;
        g[a ^ 1].push_back(b);
        g[b ^ 1].push_back(a);
    }

    // A and B
    void conjunction(int a, bool n_a, int b, bool n_b) {
        disjunction(a, n_a, a, n_a);
        disjunction(b, n_b, b, n_b);
    }

    // A -> B
    void implication(int a, bool n_a, int b, bool n_b) {
        disjunction(a, !n_a, b, n_b);
    }

    // A == B
    void equal(int a, bool n_a, int b, bool n_b) {
        implication(a, n_a, b, n_b);
        implication(b, n_b, a, n_a);
    }

    // A != B / A xor B
    void not_equal(int a, bool n_a, int b, bool n_b) {
        disjunction(a, n_a, b, n_b);
        disjunction(a, !n_a, b, !n_b);
    }

    // Solves the 2-SAT and returns a valid assignment, or empty if
        unsatisfiable
    vector<bool> build() {
        vector<vector<int>> components = tarjan(0, g);
        int m = components.size();
        vector<int> id(2 * n);
        for (int i = 0; i < m; ++i) {
            for (int a : components[i]) {
                id[a] = i;
            }
        }
        vector<bool> assignment(n);
        for (int i = 0; i < n; ++i) {
            if (id[2 * i] == id[(2 * i) ^ 1]) return {};
            assignment[i] = id[2 * i] > id[(2 * i) ^ 1];
        }
        return assignment;
    }
};
```

6.1 fast-fourier-transform

```
const float pi = acos(-1);
vector<complex<float>> root, inv_root;

void roots(int n) {
    if (n <= 2 * root.size()) return;

    root.resize(n / 2);
    inv_root.resize(n / 2);

    root[0] = {1, 0};
    inv_root[0] = {1, 0};

    float ang = 2 * pi / n;
    int half = n / 2;
    for (int i = 1; i < half; i *= 2) {
        complex<float> step(cos(ang * i), sin(ang * i));
        for (int j = 0; j < i; ++j) {
            root[i + j] = root[j] * step;
        }
    }
    for (int i = 1; i < half; ++i) {
        inv_root[i] = conj(root[i]);
    }
}

void fft(vector<complex<float>> &p, bool inv) {
    int n = p.size();
    roots(n);
    if (not inv) {
        for (int len = n; len >= 2; len /= 2) {
            int half = len / 2;
            int step = 2 * root.size() / len;
            for (int i = 0; i < n; i += len) {
                for (int j = 0; j < half; ++j) {
                    auto w = root[step * j];
                    auto x = p[i + j];
                    auto y = p[i + j + half];
                    p[i + j] = x + y;
                    p[i + j + half] = (x - y) * w;
                }
            }
        }
    } else {
        for (int len = 2; len <= n; len *= 2) {
            int half = len / 2;
            int step = 2 * root.size() / len;
            for (int i = 0; i < n; i += len) {
                for (int j = 0; j < half; ++j) {
                    auto w = inv_root[step * j];
                    auto x = p[i + j];
                    auto y = p[i + j + half] * w;
                    p[i + j] = x + y;
                    p[i + j + half] = x - y;
                }
            }
        }
        for (auto &x : p) x /= n;
    }
}

vector<int> convolution(vector<int> &a, vector<int> &b) {
    int n = a.size() + b.size() - 1;
    int m = 1;
    while (m < n) m *= 2;

    vector<complex<float>> fa(a.begin(), a.end());
    vector<complex<float>> fb(b.begin(), b.end());
    fa.resize(m);
    fb.resize(m);
    fft(fa, false);
    fft(fb, false);

    for (int i = 0; i < m; ++i) fa[i] *= fb[i];
    fft(fa, true);

    vector<int> c(n);
    for (int i = 0; i < n; ++i) c[i] = round(fa[i].real());
    return c;
}
```

6.2 fast-walsh-hadamard-transform

```
// FWHT / Bitwise Convolution: C[k] = sum(A[i] * B[j]) for i (OPR) j
    = k
// Time: O(N log N) | Space: O(N)
// Operation: 'A' (AND), 'O' (OR), 'X' (XOR)
void fwht(vector<int> &p, int sign, char opr) {
    int n = p.size();

    for (int half = 1; half < n; half *= 2) {
        for (int i = 0; i < n; i += 2 * half) {
            for (int j = 0; j < half; ++j) {
                int x = p[i + j];
                int y = p[(i + j) + half];
                if (opr == 'A') {
                    p[i + j] = ((x + sign * y) + mod) % mod;
                } else if (opr == 'O') {
                    p[(i + j) + half] = ((sign * x + y) + mod) % mod;
                } else if (opr == 'X') {
                    p[i + j] = (x + y) % mod;
                    p[(i + j) + half] = ((x - y) + mod) % mod;
                }
            }
        }
    }

    if (opr == 'X' and sign == -1) {
```

```
        int inv_n = mod_pow(n, mod - 2);
        for (int &i : p) i = (i * inv_n) % mod;
    }
}

vector<int> convolution(vector<int> a, vector<int> b, char opr) {
    int n = max(a.size(), b.size());
    int m = 1;
    while (m < n) m *= 2;

    a.resize(m);
    b.resize(m);
    fwht(a, +1, opr);
    fwht(b, +1, opr);

    for (int i = 0; i < m; ++i) a[i] = (a[i] * b[i]) % mod;
    fwht(a, -1, opr);

    return a;
}
```

6.3 modulo-transform-technique

```
vector<int> roots(int half, int ang, int mod) {
    vector<int> w(half);
    w[0] = 1;
    for (int i = 1; i < half; ++i) w[i] = (ang * w[i - 1]) % mod;
    return w;
}

void ntt(vector<int> &p, bool inv, int g, int mod) {
    int n = p.size();
    if (not inv) {
        for (int len = n; len >= 2; len /= 2) {
            int half = len / 2;
            int ang = mod_pow(g, (mod - 1) / len, mod);
            vector<int> w = roots(half, ang, mod);
            for (int i = 0; i < n; i += len) {
                for (int j = 0; j < half; ++j) {
                    int x = p[i + j];
                    int y = p[i + j + half];
                    p[i + j] = (x + y) % mod;
                    p[i + j + half] = (((x - y + mod) % mod) * w[j])
                        % mod;
                }
            }
        }
    } else {
        for (int len = 2; len <= n; len *= 2) {
            int half = len / 2;
            int ang = mod_pow(g, (mod - 1) / len, mod);
            ang = mod_pow(ang, mod - 2, mod);
            vector<int> w = roots(half, ang, mod);
            for (int i = 0; i < n; i += len) {
                for (int j = 0; j < half; ++j) {
                    int x = p[i + j];
                    int y = (p[i + j + half] * w[j]) % mod;
                    p[i + j] = (x + y) % mod;
                    p[i + j + half] = (x - y + mod) % mod;
                }
            }
        }
        int inv_n = mod_pow(n, mod - 2, mod);
        for (int &x : p) x = (x * inv_n) % mod;
    }
}

vector<int> convolution(vector<int> a, vector<int> b, int g = 3, int
    mod = 998244353) {
    int n = a.size() + b.size() - 1;
    int m = 1;
    while (m < n) m *= 2;
    a.resize(m);
    b.resize(m);
    ntt(a, false, g, mod);
    ntt(b, false, g, mod);

    for (int i = 0; i < m; ++i) a[i] = (a[i] * b[i]) % mod;
    ntt(a, true, g, mod);
    a.resize(n);

    return a;
}

vector<int> mtt(vector<int> &a, vector<int> &b, int mod) {
    vector<int> mods = {998244353, 1004535809, 469762049};

    vector<int> c1 = convolution(a, b, 3, mods[0]);
    vector<int> c2 = convolution(a, b, 3, mods[1]);
    vector<int> c3 = convolution(a, b, 3, mods[2]);

    int n = c1.size();
    vector<int> c(n);
    for (int i = 0; i < n; ++i) {
        vector<int> rems = {c1[i], c2[i], c3[i]};
        int128 x = chinese_remainder_theorem(rems, mods);
        c[i] = x % mod;
    }
    return c;
}
```

6.4 number-theoretic-transform

```
const int g = 3;
const int mod = 998244353;

vector<int> root, inv_root;

void roots(int n) {
    if (n <= 2 * root.size()) return;

    root.resize(n / 2);
    inv_root.resize( n / 2);

    root[0] = 1;
    inv_root[0] = 1;

    int ang = mod_pow(g, (mod - 1) / n);
    int inv_ang = mod_pow(ang, mod - 2);

    int half = n / 2;
    for (int i = 1; i < half; ++i) {
        root[i] = (ang * root[i - 1]) % mod;
        inv_root[i] = (inv_ang * inv_root[i - 1]) % mod;
    }
}

void ntt(vector<int> &p, bool inv) {
    int n = p.size();
    roots(n);
    if (not inv) {
        for (int len = n; len >= 2; len /= 2) {
            int half = len / 2;
            int step = 2 * root.size() / len;
            for (int i = 0; i < n; i += len) {
                for (int j = 0; j < half; ++j) {
                    int w = root[step * j];
                    int x = p[i + j];
                    int y = p[i + j + half];
                    p[i + j] = (x + y) % mod;
                    p[i + j + half] = (((x - y + mod) % mod) * w) %
                        mod;
                }
            }
        }
    } else {
        for (int len = 2; len <= n; len *= 2) {
            int half = len / 2;
            int step = 2 * root.size() / len;
            for (int i = 0; i < n; i += len) {
                for (int j = 0; j < half; ++j) {
                    int w = inv_root[step * j];
                    int x = p[i + j];
                    int y = (p[i + j + half] * w) % mod;
                    p[i + j] = (x + y) % mod;
                    p[i + j + half] = (x - y + mod) % mod;
                }
            }
        }
        int inv_n = mod_pow(n, mod - 2);
        for (int &x : p) x = (x * inv_n) % mod;
    }
}

vector<int> convolution(vector<int> a, vector<int> b) {
    int n = a.size() + b.size() - 1;
    int m = 1;
    while (m < n) m *= 2;

    a.resize(m);
    b.resize(m);
    ntt(a, false);
    ntt(b, false);

    for (int i = 0; i < m; ++i) a[i] = (a[i] * b[i]) % mod;
    ntt(a, true);
    a.resize(n);

    return a;
}
```

6.5 gaussian-elimination

```
// Gaussian Elimination: solves systems of linear equations (modulo
    arithmetic)
// Time: O(N * M^2) | Space: O(M) auxiliary
// Note: returns {(0: none, 1: unique, 2: infinite), solution}
pair<int, vector<int>> gaussian_elimination(matrix &a, const int mod)
    {
    int n = a.size(), m = a[0].size() - 1;
    vector<int> p(m, -1);
    int r, c;
    for (r = 0, c = 0; r < n and c < m; ++c) {
        int k = r;
        for (int i = r + 1; i < n and a[k][c] == 0; ++i) k = i;
        if (a[k][c] == 0) continue;

        swap(a[r], a[k]);
        p[r] = c;

        int inv = mod_pow(a[r][c], mod - 2, mod);
        for (int j = c; j <= m; ++j) {
            a[r][j] = (a[r][j] * inv) % mod;
        }

        for (int i = 0; i < n; ++i) if (i != r and a[i][c] != 0) {
            int f = mod - a[i][c];
            for (int j = c; j <= m; ++j) {
                a[i][j] = (a[i][j] + f * a[r][j]) % mod;
            }
        }
        ++r;
    }
    for (int i = r; i < n; ++i) if (a[i][m] != 0) {
        return {0, {}};
    }
    vector<int> x(m);
    for (int i = 0; i < r; ++i) {
```

```
        x[p[i]] = a[i][m];
    }
    return {(r == m) ? 1 : 2, x};
}

// Gaussian Elimination: solves systems of linear equations
     (floating-point with partial pivoting)
// Time: O(N * M^2) | Space: O(M) auxiliary
// Note: returns {(0: none, 1: unique, 2: infinite), solution}. Uses
     'eps' for numerical stability.
pair<int, vector<float>> gaussian_elimination(matrix &a, const float
     eps) {
    int n = a.size(), m = a[0].size() - 1;
    vector<int> p(m, -1);
    int r, c;
    for (r = 0, c = 0; r < n and c < m; ++c) {
        int k = r;
        for (int i = r + 1; i < n; ++i) if (abs(a[i][c]) >
             abs(a[k][c])) k = i;
        if (abs(a[k][c]) < eps) continue;

        swap(a[r], a[k]);
        p[r] = c;

        float d = a[r][c];
        for (int j = c; j <= m; ++j) {
            a[r][j] /= d;
        }

        for (int i = 0; i < n; ++i) if (i != r and abs(a[i][c]) >
             eps) {
            float f = a[i][c];
            for (int j = c; j <= m; ++j) {
                a[i][j] -= f * a[r][j];
            }
        }
        ++r;
    }
    for (int i = r; i < n; ++i) if (abs(a[i][m]) > eps) {
        return {0, {}};
    }
    vector<float> x(m);
    for (int i = 0; i < r; ++i) {
        x[p[i]] = a[i][m];
    }
    return {(r == m) ? 1 : 2, x};
}
```

6.6 josephus

```
// Josephus: finds survivor index eliminating every k-th person
// Time: O(N) | Space: O(N)
// Note: returns 0-based index
int josephus(int n, int k) {
    if (n == 1) return 0;
    return (josephus(n - 1, k) + k) % n;
}
```

6.7 berlekamp-massey

```
// Berlekamp-Massey: finds the shortest linear recurrence of a
     sequence
// Time: O(N^2) where N is sequence size | Space: O(N)
// Note: requires at least 2L terms to correctly find a recurrence of
     length L
vector<int> berlekamp_massey(const vector<int> &seq) {
    vector<int> old = {1}, curr = {1};
    int prev = 1, len = 0, shift = 1;
    for (int i = 0; i < seq.size(); ++i) {
        int delta = 0;
        for (int j = 0; j <= len; ++j) {
            delta = (delta + curr[j] * seq[i - j]) % mod;
        }
        if (delta == 0) {
            shift++;
            continue;
        }
        vector<int> temp = curr;
        int mult = (mod - (delta * power(prev, mod - 2)) % mod) % mod;
        if (curr.size() <= old.size() + shift) {
            curr.resize(old.size() + shift + 1, 0);
        }
        for (int j = 0; j < old.size(); ++j) {
            curr[j + shift] = (curr[j + shift] + mult * old[j]) % mod;
        }
        if (2 * len <= i) {
            len = i + 1 - len;
            old = temp;
            prev = delta;
            shift = 1;
        } else {
            shift++;
        }
    }
    vector<int> coef(len);
    for (int i = 1; i <= len; ++i) {
        coef[i - 1] = (mod - curr[i]) % mod;
    }
    return coef;
}
```

6.8 kth-term

```
// Linear Recurrence K-th Term: computes the K-th term of a sequence
// Time: O(L^2 log K) where L is recurrence length | Space: O(L)
// Note: 0-indexed | the sequence must have at least L terms
int kth_term(int k, const vector<int> &seq, const vector<int> &coef) {
    if (k < seq.size()) return seq[k];
    if (coef.empty()) return 0;

    int n = coef.size();

    auto combine = [&](const vector<int> &a, const vector<int> &b) ->
         vector<int> {
        vector<int> r(2 * n);
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < n; ++j) {
                r[i + j] = (r[i + j] + a[i] * b[j]) % mod;
            }
        }
        for (int i = 2 * n - 2; i >= n; --i) {
            for (int j = 0; j < n; ++j) {
                r[i - j - 1] = (r[i - j - 1] + r[i] * coef[j]) % mod;
            }
        }
        r.resize(n);
        return r;
    };

    vector<int> pol(n), exp(n);
    pol[0] = 1;
    if (n > 1) exp[1] = 1;
    else exp[0] = coef[0];

    while (k != 0) {
        if (k & 1) pol = combine(pol, exp);
        exp = combine(exp, exp);
        k >>= 1;
    }

    int ans = 0;
    for (int i = 0; i < n; ++i) {
        ans = (ans + pol[i] * seq[i]) % mod;
    }
    return ans;
}
```

6.9 matrix

```
using matrix = vector<vector<int>>;

matrix operator*(const matrix &a, const matrix &b) {
    int n = a.size();
    int m = b[0].size();
    int h = b.size();
    matrix r(n, vector<int>(m));
    for (int i = 0; i < n; ++i) {
        for (int k = 0; k < h; ++k) {
            for (int j = 0; j < m; ++j) {
                r[i][j] = (r[i][j] + a[i][k] * b[k][j]) % mod;
            }
        }
    }
    return r;
}

matrix operator^(matrix a, int b) {
    int n = a.size();
    matrix r(n, vector<int>(n));
    for (int i = 0; i < n; ++i) r[i][i] = 1;
    while (b != 0) {
        if (b & 1) r = r * a;
        a = a * a;
        b >>= 1;
    }
    return r;
}
```

6.10 max-plus-matrix

```
using matrix = vector<vector<int>>;

matrix operator*(const matrix &a, const matrix &b) {
    int n = a.size();
    int m = b[0].size();
    int h = b.size();
    matrix r(n, vector<int>(m, -inf));
    for (int i = 0; i < n; ++i) {
        for (int k = 0; k < h; ++k) {
            for (int j = 0; j < m; ++j) {
                r[i][j] = max(r[i][j], a[i][k] + b[k][j]);
            }
        }
    }
    return r;
}

matrix operator^(matrix a, int b) {
    int n = a.size();
    matrix r(n, vector<int>(n, -inf));
    for (int i = 0; i < n; ++i) r[i][i] = 0;
    while (b != 0) {
        if (b & 1) r = r * a;
        a = a * a;
        b >>= 1;
    }
    return r;
}
```

6.11 min-plus-matrix

```
using matrix = vector<vector<int>>;

matrix operator*(const matrix &a, const matrix &b) {
    int n = a.size();
    int m = b[0].size();
    int h = b.size();
    matrix r(n, vector<int>(m, inf));
    for (int i = 0; i < n; ++i) {
        for (int k = 0; k < h; ++k) {
            for (int j = 0; j < m; ++j) {
                r[i][j] = min(r[i][j], a[i][k] + b[k][j]);
```

```
            }
        }
    }
    return r;
}

matrix operator^(matrix a, int b) {
    int n = a.size();
    matrix r(n, vector<int>(n, inf));
    for (int i = 0; i < n; ++i) r[i][i] = 0;
    while (b != 0) {
        if (b & 1) r = r * a;
        a = a * a;
        b >>= 1;
    }
    return r;
}
```

6.12 nim

```
bool nim(vector<int> &piles, vector<int> &moves) {
    int n = *max(piles.begin(), piles.end());
    vector<int> grundy(n + 1);

    for (int i = 1; i <= n; ++i) {
        set<int> reach;
        for (int move : moves) if (move <= i) {
            reach.insert(grundy[i - move]);
        }

        int mex = 0;
        while (reach.contains(mex)) mex++;
        grundy[i] = mex;
    }

    int answer = 0;
    for (int pile : piles) {
        answer ^= grundy[pile];
    }
    return answer != 0;
}

bool misere(vector<int> &piles, vector<int> &moves) {
    int n = *max(piles.begin(), piles.end());
    vector<int> grundy(n + 1);

    for (int i = 1; i <= n; ++i) {
        set<int> reach;
        for (int move : moves) if (move <= i) {
            reach.insert(grundy[i - move]);
        }

        int mex = 0;
        while (reach.contains(mex)) mex++;
        grundy[i] = mex;
    }

    bool all_one = true;
    int answer = 0;
    for (int pile : piles) {
        all_one &= grundy[pile] == 1;
        answer ^= grundy[pile];
    }
    return all_one ? (piles.size() & 1) == 0 : answer != 0;
}
```

6.13 xor-basis

```
// XOR Basis: linear basis for subset XOR representation and query
    operations
// Time: insert/contains/max/kth O(log) | Space: O(log) where log is
    the maximum bit length (30)
// Note: kth(k) returns k-th smallest distinct XOR | operator[k]
    returns k-th smallest among all 2^n subsets (with duplicates)
struct xor_basis {
    static constexpr int log = 30;
    int n = 0, r = 0;
    array<int, log> basis = {};

    void insert(int x) {
        for (int i = log - 1; i >= 0; --i) if ((x >> i) & 1) {
            if (basis[i] == 0) {
                for (int j = i - 1; j >= 0; --j) if ((x >> j) & 1) {
                    x ^= basis[j];
                }
                basis[i] = x;
                r += 1;
                for (int j = log - 1; j > i; --j) if ((basis[j] >> i)
                     & 1) {
                    basis[j] ^= x;
                }
                break;
            } else {
                x ^= basis[i];
            }
        }
        n += 1;
    }

    bool contains(int x) {
        for (int i = log - 1; i >= 0; --i) if ((x >> i) & 1) {
            x ^= basis[i];
        }
        return x == 0;
    }

    int max() {
        int x = 0;
        for (int i = log - 1; i >= 0; --i) if (((x >> i) & 1) == 0) {
            x ^= basis[i];
        }
        return x;
    }

    int kth(int k) {
        if ((k >> r) != 0) return -1;
        int x = 0;
        for (int i = 0; i < log; ++i) if (basis[i] != 0) {
            if (k & 1) x ^= basis[i];
            k >>= 1;
        }
        return x;
    }

    int operator[](int k) {
        if (n - r > 60) return 0;
        return kth(k >> (n - r));
    }

    int ways(int x) {
        if (not contains(x)) return 0;
        return mod_pow(2, n - r);
    }
};
```

7.1 floor-sum

```
// Floor Sum: calculates sum{i=0}^{n-1} floor((a * i + b) / m)
// Time: O(log m) | Space: O(1)
// Note: operates similarly to the euclidean algorithm | requires n
    >= 0, m >= 1, a >= 0, b >= 0
int floor_sum(int n, int m, int a, int b) {
    int s = 0;
    while (true) {
        if (a >= m) {
            s += n * (n - 1) / 2 * (a / m);
            a %= m;
        }
        if (b >= m) {
            s += n * (b / m);
            b %= m;
        }

        int y_max = a * n + b;
        if (y_max < m) break;

        y_max /= m;
        int x_max = y_max * m - b;

        s += (n - (x_max + a - 1) / a) * y_max;
        b = (a - x_max % a) % a;

        int m_next = a;
        a = m;
        m = m_next;
        n = y_max;
    }
    return s;
}
```

7.2 harmonic-lemma

```
// Harmonic Lemma: quotient blocks in a specific range [l, r]
// Time: O(sqrt(N)) | Space: O(1)
// Note: iterates over ranges [i, j] inside [l, r] where floor(n / k)
    is exactly q
int harmonic_lemma(int n, int l, int r) {
    int s = 0;
    for (int i = l, j; i <= r; i = j + 1) {
        int q = n / i;
        if (q == 0) {
            j = r;
        } else {
            j = min(r, n / q);
        }
        s += q * (j - i + 1);
    }
    return s;
}
```

7.3 mobius

```
int mobius(int n) {
    int mu = 1;
    for (int i = 2; i * i <= n; ++i) if (n % i == 0) {
        n /= i;
        if (n % i == 0) return 0;
        mu *= -1;
    }
    if (n > 1) mu *= -1;
    return mu;
}

vector<int> mobius_sieve(int n) {
    vector<int> mu(n + 1);
    vector<bool> is_prime(n + 1, true);
    vector<int> primes;
    mu[1] = 1;
    is_prime[1] = false;
    for (int i = 2; i <= n; ++i) {
        if (is_prime[i]) {
            primes.push_back(i);
            mu[i] = -1;
        }
        for (int p : primes) {
            if (i * p > n) break;
            is_prime[i * p] = false;
            if (i % p == 0) {
                mu[i * p] = 0;
                break;
            } else {
```

```
                mu[i * p] = -mu[i];
            }
        }
    }
    return mu;
}
```

7.4 chinese-remainder-theorem

```
int128 chinese_remainder_theorem(const vector<int> &rem, const
     vector<int> &mod) {
    int128 r = rem[0], m = mod[0];
    for (int i = 1; i < rem.size(); ++i) {
        int g = gcd(static_cast<int>(m % mod[i]), mod[i]), h = mod[i]
             / g;
        if ((rem[i] - r) % g != 0) return -1;

        auto [x, _] = extended_gcd(static_cast<int>(m / g % h), h);

        r += m * (((((rem[i] - r) / g) * x) % h + h) % h);
        m *= h;
    }
    return r;
}
```

7.5 extended-gcd

```
// Extended GCD: finds coefficients x and y such that ax + by =
     gcd(a, b)
// Time: O(log(min(a, b))) | Space: O(log(min(a, b)))
// Note: returns [x0, y0] | all valid pairs are: x = x0 + k*(b/gcd)
     and y = y0 - k*(a/gcd)
pair<int, int> extended_gcd(int a, int b) {
    if (b == 0) return {1, 0};

    auto [x, y] = extended_gcd(b, a % b);
    return {y, x - (a / b) * y};
}
```

7.6 mod-log

```
int mod_log(int a, int b, int mod) {
    if (a == 0) {
        if (b == 0) return 1 % mod;
        return -1;
    }

    int k = 1, h = 0;
    for (int g = gcd(a, mod); g != 1; g = gcd(a, mod), ++h) {
        if (k == b) return h;
        if (b % g) return -1;

        b /= g, mod /= g;
        k = (k * (a / g)) % mod;
    }

    int n = ceil(sqrt(mod)), s = 1;
    for (int i = 1; i <= n; ++i) {
        s = (s * a) % mod;
    }

    map<int, int> f;
    for (int q = 0; q <= n; ++q) {
        f[b] = q;
        b = (b * a) % mod;
    }

    a = (k * s) % mod;
    for (int p = 1; p <= n; ++p) {
        if (f.contains(a)) {
            return (n * p - f[a]) + h;
        }
        a = (a * s) % mod;
    }
    return -1;
}
```

7.7 mod-pow

```
int mod_pow(int a, int e, const int mod) {
    int r = 1;
    while (e != 0) {
        if (e & 1) r = (r * a) % mod;
        a = (a * a) % mod;
        e >>= 1;
    }
    return r;
}
```

7.8 multi-extended-gcd

```
vector<int> multi_extended_gcd(vector<int> &a) {
    int n = a.size(), g = a[0];
    vector<int> c(n, 1), h(n - 1);

    for (int i = 1; i < n; ++i) {
        auto [x, y] = extended_gcd(g, a[i]);
        g = gcd(g, a[i]);
        h[i - 1] = x;
        c[i] = y;
    }

    int p = 1;
    for (int i = n - 1; i > 0; --i) {
        c[i] *= p;
        p *= h[i - 1];
    }
    c[0] *= p;
    return c;
}
```

7.9 linear-sieve

```
vector<int> linear_sieve(int n) {
    vector<int> p, spf(n + 1);
    for (int i = 2; i <= n; ++i) {
        if (spf[i] == 0) {
            spf[i] = i;
            p.push_back(i);
        }
        for (int j : p) {
            if (j > spf[i] or i * j > n) break;
            spf[i * j] = j;
        }
    }
    return p;
}
```

7.10 miller-rabin

```
int mod_pow(int128 a, int e, const int mod) {
    int128 r = 1;
    while (e != 0) {
        if (e & 1) r = (r * a) % mod;
        a = (a * a) % mod;
        e >>= 1;
    }
    return r;
}

bool composite(int n, int a, int s, int d) {
    int128 x = mod_pow(a, d, n);
    if (x == 1 or x == n - 1) return false;
    for (int i = 1; i < s; ++i) {
        x = (x * x) % n;
        if (x == n - 1) return false;
        if (x == 1) return true;
    }
    return true;
}

bool miller_rabin(int n) {
    if (n <= 1) return false;
    static const int P[] = {2, 3, 5, 13, 19, 73, 193, 407521, 29921083
         7};
    for (int p : P) {
        if (n == p) return true;
        if (n % p == 0) return false;
    }
    int s = __builtin_ctzll(n - 1);
    int d = (n - 1) >> s;
    static const int A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795
          265022};
    for (int a : A) {
        if (composite(n, a, s, d)) return false;
    }
    return true;
}
```

7.11 pollard-rho

```
// Pollard-Rho: fast integer factorization
// Time: O(N^(1/4)) expected | Space: O(1)
// Note: finds a non-trivial divisor | use recursively to fully factor
mt19937_64
     rng(chrono::steady_clock::now().time_since_epoch().count());

int f(int128 x, int c, const int mod) {
    return (x * x + c) % mod;
}

int pollard_rho(int n) {
    if ((n & 1) == 0) return 2;
    if (miller_rabin(n)) return n;
    int c = uniform_int_distribution<int>(1, n - 1)(rng);
    int x = 2, y = 2, g = 1, l = 1, p = 2, q = 1;
    while (g == 1) {
        y = x;
        int k = 0;
        while (k < l and g == 1) {
            p = x;
            for (int i = 1; i <= min<int>(128, l - k); ++i) {
                x = f(x, c, n);
                q = ((int128) q * abs(x - y)) % n;
            }
            g = gcd(q, n);
            k += 128;
        }
        l *= 2;
    }
    if (g == n) {
        do {
            p = f(p, c, n);
            g = gcd(abs(p - y), n);
        } while (g == 1);
    }
    if (g == n) return pollard_rho(n);
    return g;
}
```

7.12 segmented-sieve

```
vector<int> segmented_sieve(int l, int r) {
    vector<int> primes = linear_sieve(sqrt(r));
    offset_array<char> is_prime(l, r + 1, true);
    for (int p : primes) {
        for (int i = max(p * ((l + p - 1) / p), p * p); i <= r; i +=
             p) {
            is_prime[i] = false;
        }
    }
    primes.clear();
    for (int i = max(2, l); i <= r; ++i) {
        if (is_prime[i]) {
            primes.push_back(i);
```

```
        }
    }
    return primes;
}
```

7.13 totient

```
int totient(int n) {
    int phi = n;
    for (int i = 2; i * i <= n; ++i) if (n % i == 0) {
        phi -= phi / i;
        while (n % i == 0) n /= i;
    }
    if (n > 1) phi -= phi / n;
    return phi;
}

vector<int> totient_sieve(int n) {
    vector<int> phi(n + 1);
    for (int i = 1; i <= n; ++i) {
        phi[i] = i;
    }
    for (int i = 2; i <= n; ++i) if (phi[i] == i) {
        for (int j = i; j <= n; j += i) {
            phi[j] -= phi[j] / i;
        }
    }
    return phi;
}
```

8.1 decomposition

```
struct decomposition {
    int n, m;
    vector<int> a, b;

    decomposition(vector<int> &base, float f = 1.0) {
        n = base.size();
        m = clamp<int>(f * sqrt(n), 1, n);
        a.resize(n);
        b.resize((n + (m - 1)) / m);
        for (int i = 0; i < n; ++i) {
            a[i] = base[i];
            b[i / m] += a[i];
        }
    }

    void update(int i, int x) {
        a[i] = x;
        int b_i = i / m;
        b[b_i] = 0;
        int l = b_i * m;
        int r = min(n, (b_i + 1) * m) - 1;
        for (int k = l; k <= r; ++k) {
            b[b_i] += a[k];
        }
    }

    int query(int l, int r) {
        int s = 0;
        int b_l = l / m;
        int b_r = r / m;
        if (b_l == b_r) {
            for (int i = l; i <= r; ++i) {
                s += a[i];
            }
        } else {
            for (int i = l; i < (b_l + 1) * m; ++i) {
                s += a[i];
            }
            for (int i = b_l + 1; i < b_r; ++i) {
                s += b[i];
            }
            for (int i = b_r * m; i <= r; ++i) {
                s += a[i];
            }
        }
        return s;
    }
};
```

8.2 mo-3d

```
// Mo's Algorithm 3D: offline range queries with point updates
// Time: O(N^(5/3)) | Space: O(N + Q + U)
// Q elements: {l, r, t, id} | t = # of updates prior to query | id =
    original query index
// U elements: {i, x} | i = array index to update | x = new value
// M parameter: block size multiplier to tune the N^(2/3) block size
// Note: implement 'insert', 'remove', and 'curr' state according to
    your specific problem
vector<int> mo_3d(vector<int> &a, vector<array<int, 4>> &q,
     vector<array<int, 2>> &u, float m = 1.0) {
    int b = clamp<int>(m * pow(a.size(), 2.0 / 3.0), 1, a.size());
    auto block = [&](int i) -> int {
        return i / b;
    };

    sort(q.begin(), q.end(), [&](const array<int, 4> &x, const
         array<int, 4> &y) {
        if (block(x[0]) != block(y[0])) return block(x[0]) <
             block(y[0]);
        if (block(x[1]) != block(y[1])) return (block(x[0]) & 1) ?
             block(x[1]) < block(y[1]) : block(x[1]) > block(y[1]);
        return (block(x[1]) & 1) ? x[2] < y[2] : x[2] > y[2];
    });

    int curr = 0;
    auto insert = [&](int x) -> void {};
    auto remove = [&](int x) -> void {};

    auto update = [&](int t, int l, int r) -> void {
        int i = u[t][0];
        if (l <= i and i <= r) {
            remove(a[i]);
            swap(a[i], u[t][1]);
            insert(a[i]);
        } else {
            swap(a[i], u[t][1]);
        }
    };

    vector<int> ans(q.size());
    int h = 0, i = 0, j = -1;
    for (auto &[l, r, t, id] : q) {
        while (h < t) update(h++, i, j);
        while (h > t) update(--h, i, j);
        while (i > l) insert(a[--i]);
        while (j < r) insert(a[++j]);
        while (i < l) remove(a[i++]);
        while (j > r) remove(a[j--]);
        ans[id] = curr;
    }
    return ans;
}
```

8.3 mo

```
int hilbert(int x, int y) {
    int h = 0;
    for (int s = 25; s >= 0; --s) {
        int rx = (x >> s) & 1;
        int ry = (y >> s) & 1;
        int qd = (3 * rx) ^ ry;
        h = (h << 2) | qd;
        if (ry == 0) {
            x ^= -rx;
            y ^= -rx;
            swap(x, y);
        }
    }
    return h;
}

vector<int> mo(vector<int> &a, vector<array<int, 4>> &q) {
    for (auto &[h, l, r, id] : q) h = hilbert(l, r);
    sort(q.begin(), q.end());

    int curr = 0;
    auto insert = [&](int x) -> void {};
    auto remove = [&](int x) -> void {};

    vector<int> ans(q.size());
    int i = 0, j = -1;
    for (auto &[h, l, r, id] : q) {
        while (i > l) insert(a[--i]);
        while (j < r) insert(a[++j]);
        while (i < l) remove(a[i++]);
        while (j > r) remove(a[j--]);
        ans[id] = curr;
    }
    return ans;
}
```

9.1 aho-corasick

```
struct aho_corasick {
    struct node {
        int fail = 0;
        int exit = 0;
        int len = 0;
        int cnt = 0;
        array<int, 26> next = {};
    };

    vector<node> trie;

    aho_corasick(int n = 0) {
        trie.reserve(n + 5);
        trie.emplace_back();
    }

    void insert(string &s) {
        int i = 0;
        for (char c : s) {
            int k = c - 'a';
            if (trie[i].next[k] == 0) {
                trie[i].next[k] = trie.size();
                trie.emplace_back();
            }
            i = trie[i].next[k];
        }
        trie[i].len = s.length();
        trie[i].cnt++;
    }

    void compile() {
        queue<int> q;
        for (int c = 0; c < 26; ++c) {
            int i = trie[0].next[c];
            if (i != 0) q.push(i);
        }
        while (q.size()) {
            int i = q.front(); q.pop();
            int f = trie[i].fail;
            trie[i].exit = trie[f].cnt != 0 ? f : trie[f].exit;
            for (int c = 0; c < 26; ++c) {
                int &j = trie[i].next[c];
                if (j != 0) {
                    trie[j].fail = trie[f].next[c];
                    q.push(j);
```

```
                } else {
                    j = trie[f].next[c];
                }
            }
        }
    }

    vector<int> match(string &t) {
        int n = t.length();
        vector<int> p;
        int i = 0;
        for (int k = 0; k < n; ++k) {
            i = trie[i].next[t[k] - 'a'];
            for (int j = i; j != 0; j = trie[j].exit) if (trie[j].cnt
                 != 0) {
                for (int r = 1; r <= trie[j].cnt; ++r) {
                    p.push_back(k - trie[j].len + 1);
                }
            }
        }
        return p;
    }
};
```

9.2 deterministic-finite-automaton

```
vector<array<int, 26>> deterministic_finite_automaton(const string
     &s) {
    int n = s.length();
    vector<array<int, 26>> dfa(n + 1);
    if (n == 0) return dfa;
    dfa[0][s[0] - 'a'] = 1;
    for (int i = 1, j = 0; i <= n; ++i) {
        for (int c = 0; c < 26; ++c) {
            dfa[i][c] = dfa[j][c];
        }
        if (i < n) {
            dfa[i][s[i] - 'a'] = i + 1;
            j = dfa[j][s[i] - 'a'];
        }
    }
    return dfa;
}
```

9.3 dynamic-hasher

```
block init(char c) {
    block h;
    h.len = 1;
    for (int i = 0; i <= 1; ++i) {
        h.ord[i] = c % mod[i];
        h.rev[i] = c % mod[i];
    }
    return h;
}

block combine(const block &l, const block &r) {
    if (l.len == 0) return r;
    if (r.len == 0) return l;
    extend(max(l.len, r.len));
    block m;
    m.len = l.len + r.len;
    for (int i = 0; i <= 1; ++i) {
        m.ord[i] = (l.ord[i] * pows[r.len][i] + r.ord[i]) % mod[i];
        m.rev[i] = (r.rev[i] * pows[l.len][i] + l.rev[i]) % mod[i];
    }
    return m;
}

struct dynamic_hasher {
    int n;
    vector<block> tree;

    dynamic_hasher(const string &s): n(s.length()), tree(2 * n) {
        for (int i = 0; i < n; ++i) tree[n + i] = init(s[i]);
        for (int i = n - 1; i > 0; --i) tree[i] = combine(tree[i << 1
             ], tree[(i << 1) | 1]);
    }

    void update(int i, char c) {
        tree[i += n] = init(c);
        for (i >>= 1; i > 0; i >>= 1) tree[i] = combine(tree[i << 1],
             tree[(i << 1) | 1]);
    }

    block query(int i, int j) {
        block l, r;
        for (i += n, j += n + 1; i < j; i >>= 1, j >>= 1) {
            if (i & 1) l = combine(l, tree[i++]);
            if (j & 1) r = combine(tree[--j], r);
        }
        return combine(l, r);
    }
};
```

9.4 hasher

```
const array<int, 2> base = {41, 53};
const array<int, 2> mod = {1000000007, 1000000009};

vector<array<int, 2>> pows = {{{1, 1}}};

void extend(int n) {
    while (pows.size() <= n) {
        array<int, 2> back = pows.back();
        array<int, 2> curr = {(back[0] * base[0]) % mod[0], (back[1]
             * base[1]) % mod[1]};
        pows.push_back(curr);
    }
}

struct block {
    int len = 0;
    array<int, 2> ord = {};
    array<int, 2> rev = {};

    bool operator==(const block &o) const {
        return len == o.len and ord == o.ord;
    }
};

struct hasher {
    int n;
    vector<array<int, 2>> pref, suf;

    hasher(const string &s): n(s.length()), pref(n + 1), suf(n + 2) {
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j <= 1; ++j) {
                pref[i + 1][j] = (pref[i][j] * base[j] + s[i]) %
                     mod[j];
                suf[n - i][j] = (suf[n - i + 1][j] * base[j] + s[n -
                     i - 1]) % mod[j];
            }
        }
        extend(n);
    }

    block query(int l, int r) const {
        block h;
        h.len = r - l + 1;
        for (int i = 0; i <= 1; ++i) {
            h.ord[i] = (pref[r + 1][i] - pref[l][i] * pows[h.len][i])
                 % mod[i];
            h.ord[i] = (h.ord[i] + mod[i]) % mod[i];
            h.rev[i] = (suf[l + 1][i] - suf[r + 2][i] *
                 pows[h.len][i]) % mod[i];
            h.rev[i] = (h.rev[i] + mod[i]) % mod[i];
        }
        return h;
    }
};
```

9.5 longest-prefix-suffix

```
vector<int> longest_prefix_suffix(string &s) {
    int n = s.length();
    vector<int> lps(n);
    for (int i = 1; i < n; ++i) {
        int j = lps[i - 1];
        while (j != 0 and s[i] != s[j]) j = lps[j - 1];
        if (s[i] == s[j]) j++;
        lps[i] = j;
    }
    return lps;
}
```

9.6 manacher

```
string pre_process(string &s) {
    string t = "#";
    for (char &c : s) {
        t.push_back(c);
        t.push_back('#');
    }
    return t;
}

vector<int> manacher(string &s) {
    string t = '^' + pre_process(s) + '$';
    int n = t.length();

    vector<int> p(n);
    int c = 0, r = 0;
    for (int i = 1; i < n - 1; i++) {
        int j = 2 * c - i;
        if (i < r) {
            p[i] = min(r - i, p[j]);
        }
        while (t[i + (p[i] + 1)] == t[i - (p[i] + 1)]) {
            p[i]++;
        }
        if (i + p[i] > r) {
            c = i;
            r = i + p[i];
        }
    }
    return p;
}

string longest_palindrome(string &s) {
    vector<int> p = manacher(s);
    int n = p.size();

    int k = 0, l = 0;
    for (int i = 1; i < n - 1; i++) {
        if (p[i] > l) {
            l = p[i];
            k = (i - l) / 2;
        }
    }
    return s.substr(k, l);
}

int count_palindromes(string &s) {
    vector<int> p = manacher(s);
    int n = p.size();

    int total = 0;
    for (int i = 1; i < n - 1; ++i) {
        total += (p[i] + 1) / 2;
    }
```

```
    return total;
}
```

9.7 palindromic-tree

```
struct palindromic_tree {
    struct node {
        int length, fail;
        array<int, 26> child = {};

        node(int length): length(length), fail(0) {}
    };

    int suffix;
    vector<char> text;
    vector<node> tree;

    palindromic_tree(int n): suffix(0) {
        text.reserve(n + 5);
        text.push_back('$');
        tree.reserve(n + 5);
        tree.push_back(node(0));
        tree.push_back(node(-1));
        tree[0].fail = 1;
    }

    int follow(int id) {
        while (text[(text.size() - tree[id].length) - 2] !=
             text.back()) {
            id = tree[id].fail;
        }
        return id;
    }

    void insert(char c) {
        text.push_back(c);
        suffix = follow(suffix);

        if (not tree[suffix].child[c - 'a']) {
            int id = tree.size();
            tree.push_back(node(tree[suffix].length + 2));

            tree[id].fail = tree[follow(tree[suffix].fail)].child[c -
                 'a'];
            tree[suffix].child[c - 'a'] = id;
        }
        suffix = tree[suffix].child[c - 'a'];
    }
};
```

9.8 suffix-array

```
// Suffix Array: lexicographical sorting of all string suffixes
// Time: build O(N log N) | LCP O(N) | search O(|P| log N) | Space:
    O(N)
// Note: expects sentinel character at the end of input | supports
    pattern matching and k-th queries
struct suffix_array {
    int n, m;
    string s;
    vector<int> suf, lcp;
    vector<int> rank, temp;

    void sort() {
        vector<int> f(m + 1);
        for (int i = 0; i < n; ++i) {
            int r = rank[suf[i]];
            f[r] += 1;
        }
        for (int i = 1; i <= m; ++i) {
            f[i] += f[i - 1];
        }
        for (int i = n - 1; i >= 0; --i) {
            int r = rank[suf[i]];
            f[r] -= 1;
            temp[f[r]] = suf[i];
        }
        swap(suf, temp);
    }

    suffix_array(string &s): n(s.length()), m(0), s(s), suf(n),
         lcp(n), rank(n), temp(n) {
        for (int i = 0; i < n; ++i) {
            suf[i] = i;
            rank[i] = s[i];
            m = max<int>(m, s[i]);
        }
        sort();
        for (int k = 1; k < n; k <<= 1) {
            for (int i = 0; i < n; ++i) {
                suf[i] = (suf[i] - k + n) % n;
            }
            sort();
            temp[suf[0]] = 1;
            for (int i = 1; i < n; ++i) {
                int prev = suf[i - 1];
                int curr = suf[i];
                if (rank[prev] == rank[curr] and rank[(prev + k) % n]
                     == rank[(curr + k) % n]) {
                    temp[curr] = temp[prev];
                } else {
                    temp[curr] = temp[prev] + 1;
                }
            }
            swap(rank, temp);
            m = rank[suf[n - 1]];
            if (m == n) break;
        }
    }

    void longest_common_prefix() {
        vector<int> inv(n);
        for (int i = 0; i < n; ++i) {
            inv[suf[i]] = i;
        }
        int l = 0;
        for (int i = 0; i < n; ++i) {
            if (inv[i] == 0) {
                l = 0;
                continue;
            }
            int j = suf[inv[i] - 1];
            while (i + l < n and j + l < n and s[i + l] == s[j + l])
                 l++;
            lcp[inv[i]] = l;
            l = max<int>(0, l - 1);
        }
    }

    int compare(int i, string &t) {
        return s.compare(i, t.size(), t);
    }

    int lower_search(string &t) {
        int answer = 0;
        int low = 1, high = n - 1;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (compare(suf[mid], t) < 0) {
                low = mid + 1;
            } else {
                answer = mid;
                high = mid - 1;
            }
        }
        return answer;
    }

    int upper_search(string &t) {
        int answer = n;
        int low = 1, high = n - 1;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (compare(suf[mid], t) <= 0) {
                low = mid + 1;
            } else {
                answer = mid;
                high = mid - 1;
            }
        }
        return answer;
    }

    int search(string &t) {
        int lower = lower_search(t);
        if (compare(suf[lower], t) != 0) return 0;
        int upper = upper_search(t);
        return upper - lower;
    }

    pair<int, int> kth_distinct_substring(int k) {
        assert(k >= 0);
        for (int i = 1; i < n; ++i) {
            int len = (n - 1) - suf[i];
            int sub = len - lcp[i];
            if (k < sub) return {suf[i], lcp[i] + k + 1};
            k -= sub;
        }
        return {-1, -1};
    }

    pair<int, int> kth_substring(int k) {
        assert(k >= 0);
        sparse_table st(lcp);
        for (int i = 1; i < n; ++i) {
            int curr_len = lcp[i] + 1;
            int max_len = (n - 1) - suf[i];
            while (curr_len <= max_len) {
                int j = i;
                int low = i + 1, high = n - 1;
                while (low <= high) {
                    int mid = (low + high) / 2;
                    if (st.query(i + 1, mid) >= curr_len) {
                        j = mid;
                        low = mid + 1;
                    } else {
                        high = mid - 1;
                    }
                }
                int freq = j - i + 1;
                int len = max_len;
                if (j > i) len = min(len, st.query(i + 1, j));
                int sub = (len - curr_len + 1) * freq;
                if (k < sub) return {suf[i], curr_len + (k / freq)};
                k -= sub;
                curr_len = len + 1;
            }
        }
        return {-1, -1};
    }
};
```

9.9 suffix-automaton

```
// Suffix Automaton: directed acyclic graph representing all
    substrings of a string
// Time: extend O(1) amortized | Space: O(N * S)
// Note: root at index 1 | alphabet [a-z]
struct suffix_automaton {
    struct node {
```

```
        int len = 0, link = 0;
        int pos = -1, cnt = 0;
        array<int, 26> next = {};
    };

    int prev;
    vector<node> st;

    suffix_automaton(): prev(1), st(2) {}

    void extend(char c) {
        int idx = c - 'a';
        int curr = st.size();
        int ptr = prev;

        st.emplace_back();
        st[curr].len = st[prev].len + 1;
        st[curr].pos = st[curr].len - 1;
        st[curr].cnt = 1;
        prev = curr;

        while (ptr and (not st[ptr].next[idx])) {
            st[ptr].next[idx] = curr;
            ptr = st[ptr].link;
        }

        if (not ptr) {
            st[curr].link = 1;
            return;
        }

        int qtr = st[ptr].next[idx];
        if (st[ptr].len + 1 == st[qtr].len) {
            st[curr].link = qtr;
            return;
        }

        int clone = st.size();
        st.push_back(st[qtr]);
        st[clone].len = st[ptr].len + 1;
        st[clone].cnt = 0;

        while (ptr and st[ptr].next[idx] == qtr) {
            st[ptr].next[idx] = clone;
            ptr = st[ptr].link;
        }

        st[curr].link = st[qtr].link = clone;
    }

    bool contains(string &t) {
        int ptr = 1;
        for (char c : t) {
            int idx = c - 'a';
            ptr = st[ptr].next[idx];
            if (not ptr) return false;
        }
        return true;
    }
};
```

9.10 tree-suffix-array

```
// Tree Suffix Array: lexicographical sorting of all tree paths (node
    to root)
// Time: build O(N log N) | LCP O(log N) | search O(|P| log N) |
    Space: O(N log N)
// Note: 1-indexed nodes, node 0 is reserved as a sentinel root with
    s[0] = '$'
struct tree_suffix_array {
    int n, m, log;
    string s;
    vector<int> suf, lcp, temp, depth, freq;
    vector<vector<int>> up, rank;

    void dfs(int a, int p, int d, vector<vector<int>> &g) {
        up[0][a] = p;
        depth[a] = d;
        for (int k = 1; k <= log; ++k) {
            up[k][a] = up[k - 1][up[k - 1][a]];
        }
        for (int b : g[a]) if (b != p) {
            dfs(b, a, d + 1, g);
        }
    }

    void sort(int k, bool jump) {
        fill(freq.begin(), freq.begin() + m + 1, 0);
        for (int i = 0; i < n; ++i) {
            int h = suf[i];
            if (jump) h = up[k][h];
            int r = rank[k][h];
            freq[r]++;
        }
        for (int i = 1; i <= m; ++i) {
            freq[i] += freq[i - 1];
        }
        for (int i = n - 1; i >= 0; --i) {
            int h = suf[i];
            if (jump) h = up[k][h];
            int r = rank[k][h];
            freq[r]--;
            temp[freq[r]] = suf[i];
        }
        swap(suf, temp);
    }

    tree_suffix_array(const string &letters, vector<vector<int>> &g):
         n(letters.length()), m(0), log(__lg(n | 1) + 1),
    s(letters), suf(n), lcp(n), temp(n), depth(n), freq(max<int>(256,
         n)), up(log + 1, vector<int>(n)), rank(log + 1,
         vector<int>(n)) {
        dfs(1, 0, 1, g);
        for (int i = 0; i < n; ++i) {
            suf[i] = i;
            rank[0][i] = s[i];
            m = max<int>(m, s[i]);
        }
        for (int k = 0; k < log; ++k) {
            sort(k, true);
            sort(k, false);
            rank[k + 1][suf[0]] = 0;
            int val = 0;
            for (int i = 1; i < n; ++i) {
                int prev = suf[i - 1];
                int curr = suf[i];
                int j_prev = up[k][prev];
                int j_curr = up[k][curr];
                if (rank[k][prev] == rank[k][curr] and
                     rank[k][j_prev] == rank[k][j_curr]) {
                    rank[k + 1][curr] = val;
                } else {
                    rank[k + 1][curr] = ++val;
                }
            }
            m = val;
            if (m == n - 1) {
                for (int h = k + 1; h <= log; ++h) rank[h] = rank[k +
                     1];
                break;
            }
        }
    }

    int longest_common_prefix(int a, int b) {
        if (a == b) return depth[a];
        int answer = 0;
        for (int k = log; k >= 0 and a != 0 and b != 0; --k) if
             (rank[k][a] == rank[k][b]) {
            answer += 1 << k;
            a = up[k][a];
            b = up[k][b];
        }
        return answer;
    }

    void build_longest_common_prefix() {
        for (int i = 1; i < n; ++i) {
            lcp[i] = longest_common_prefix(suf[i], suf[i - 1]);
        }
    }

    int compare(int a, string &t) {
        for (int i = 0; i < t.length(); ++i) {
            if (a == 0) return -1;
            if (s[a] != t[i]) return s[a] < t[i] ? -1 : +1;
            a = up[0][a];
        }
        return 0;
    }

    int lower_search(string &t) {
        int answer = 0;
        int low = 1, high = n - 1;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (compare(suf[mid], t) < 0) {
                low = mid + 1;
            } else {
                answer = mid;
                high = mid - 1;
            }
        }
        return answer;
    }

    int upper_search(string &t) {
        int answer = n;
        int low = 1, high = n - 1;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (compare(suf[mid], t) <= 0) {
                low = mid + 1;
            } else {
                answer = mid;
                high = mid - 1;
            }
        }
        return answer;
    }

    int search(string &t) {
        int lower = lower_search(t);
        if (compare(suf[lower], t) != 0) return 0;
        int upper = upper_search(t);
        return upper - lower;
    }
};
```

9.11 trie

```
struct trie {
    struct node {
        int pref = 0;
        int end = 0;
        array<int, 26> child = {};
    };

    vector<node> tree;
```

```
    trie(int n = 0) {
        tree.reserve(n + 1);
        tree.emplace_back();
    }

    void insert(string &s) {
        int i = 0;
        tree[0].pref++;
        for (char c : s) {
            int k = c - 'a';
            if (tree[i].child[k] == 0) {
                tree[i].child[k] = tree.size();
                tree.emplace_back();
            }
            i = tree[i].child[k];
            tree[i].pref++;
        }
        tree[i].end++;
    }

    void remove(string &s) {
        if (not contains(s)) return;
        int i = 0;
        tree[i].pref--;
        for (char c : s) {
            int k = c - 'a';
            i = tree[i].child[k];
            tree[i].pref--;
        }
        tree[i].end--;
    }

    bool contains(string &s) {
        int i = 0;
        for (char c : s) {
            int k = c - 'a';
            i = tree[i].child[k];
            if (i == 0) return false;
        }
        return tree[i].end != 0;
    }
};
```

9.12 z-array

```
vector<int> z_array(string &s) {
    int n = s.length();
    vector<int> z(n);
    for (int i = 1, l = 0, r = 0; i < n; ++i) {
        z[i] = max(0, min(z[i - l], r - (i + 1)));
        while (i + z[i] < n and s[i + z[i]] == s[z[i]]) {
            l = i;
            r = i + z[i];
            z[i] += 1;
        }
    }
    return z;
}
```

10.1 binary-indexed-tree-2d

```
struct bit_2d {
    int n, m;
    vector<bit> tree;

    bit_2d(int n = 0, int m = 0): n(n), m(m), tree(n + 1, bit(m)) {}

    void build(vector<vector<int>> &a) {
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < m; ++j) {
                update(i, j, a[i][j]);
            }
        }
    }

    void update(int i, int j, int x) {
        for (i += 1; i <= n; i += i & -i) tree[i].update(j, x);
    }

    int prefix(int i, int j) {
        int s = 0;
        for (i += 1; i > 0; i -= i & -i) s += tree[i].prefix(j);
        return s;
    }

    int query(int i, int j, int p, int q) {
        return prefix(p, q)
             - prefix(i - 1, q)
             - prefix(p, j - 1)
             + prefix(i - 1, j - 1);
    }
};
```

10.2 binary-indexed-tree

```
struct bit {
    int n;
    vector<int> tree;

    static int lsb(int i) {
        return i & -i;
    }

    bit(int n = 0): n(n), tree(n + 1) {}

    void update(int i, int x) {
        for (i += 1; i <= n; i += lsb(i)) tree[i] += x;
    }

    int prefix(int i) {
        int s = 0;
        for (i += 1; i > 0; i -= lsb(i)) s += tree[i];
        return s;
    }

    int query(int i, int j) {
        return prefix(j) - prefix(i - 1);
    }

    int find_kth(int k) {
        k += 1;
        assert(k > 0);
        int i = 0;
        for (int s = 1 << __lg(n); s > 0; s >>= 1) if (i + s <= n and
             tree[i + s] < k) {
            i += s;
            k -= tree[i];
        }
        return i;
    }
};
```

10.3 range-binary-indexed-tree

```
struct range_bit {
    bit m, s;

    range_bit(int n): m(n + 1), s(n + 1) {}

    void update(int i, int j, int x) {
        m.update(i, x);
        m.update(j + 1, -x);
        s.update(i, x * (i - 1));
        s.update(j + 1, -x * j);
    }

    int prefix(int i) {
        return i * m.prefix(i) - s.prefix(i);
    }

    int query(int i, int j) {
        return prefix(j) - prefix(i - 1);
    }
};
```

10.4 sparse-binary-indexed-tree

```
// Sparse BIT: 2D Binary Indexed Tree with Static Coordinate
     Compression
// Time: Build O(Q log^2 Q) | Update/Query O(log^2 Q) | Space: O(Q
     log Q)
// Queries: {type, i1, j1, i2, j2} | 'U' = Update, 'Q' = Query
struct sparse_bit {
    int n;
    vector<int> is;
    vector<vector<int>> js;
    vector<bit> tree;

    void compress(vector<int> &a) {
        sort(a.begin(), a.end());
        a.erase(unique(a.begin(), a.end()), a.end());
    }

    int id(vector<int> &a, int i) {
        return lower_bound(a.begin(), a.end(), i) - a.begin();
    }

    sparse_bit(vector<tuple<char, int, int, int, int>> &q) {
        for (auto [t, i1, j1, i2, j2] : q) {
            if (t == 'U') is.push_back(i1);
            else if (t == 'Q') is.push_back(i1 - 1), is.push_back(i2);
        }
        compress(is);
        n = is.size();

        js.resize(n + 1);
        for (auto [t, i1, j1, i2, j2] : q) {
            if (t == 'U') {
                for (i1 = id(is, i1) + 1; i1 <= n; i1 +=
                     bit::lsb(i1)) js[i1].push_back(j1);
            } else if (t == 'Q') {
                for (i1 = id(is, i1 - 1) + 1; i1 > 0; i1 -=
                     bit::lsb(i1)) js[i1].push_back(j1 - 1),
                     js[i1].push_back(j2);
                for (i2 = id(is, i2) + 1; i2 > 0; i2 -= bit::lsb(i2))
                     js[i2].push_back(j1 - 1), js[i2].push_back(j2);
            }
        }

        tree.resize(n + 1);
        for (int i = 1; i <= n; ++i) {
            compress(js[i]);
            tree[i] = bit(js[i].size());
        }
    }

    void update(int i, int j, int x) {
        for (i = id(is, i) + 1; i <= n; i += bit::lsb(i))
             tree[i].update(id(js[i], j), x);
    }

    int prefix(int i, int j) {
        int s = 0;
        for (i = id(is, i) + 1; i > 0; i -= bit::lsb(i)) s +=
             tree[i].prefix(id(js[i], j));
        return s;
    }

    int query(int i1, int j1, int i2, int j2) {
        return prefix(i2, j2) - prefix(i1 - 1, j2) - prefix(i2, j1 - 1
```

```
            ) + prefix(i1 - 1, j1 - 1);
    }
};
```

10.5 cover-tree

```
// Cover Tree: calculates total length covered by at least one
    interval
// Time: update O(log N), query O(1) | Space: O(N)
struct cover_tree {
    int n;
    vector<int> all, cnt, curr;

    void recalc(int i) {
        if (cnt[i] > 0) curr[i] = all[i];
        else if (i < n) curr[i] = curr[i << 1] + curr[i << 1 | 1];
        else curr[i] = 0;
    }

    cover_tree(vector<int> &x): n(x.size() - 1), all(2 * n), cnt(2 *
        n), curr(2 * n) {
        for (int i = 0; i < n; ++i) all[n + i] = x[i + 1] - x[i];
        for (int i = n - 1; i > 0; --i) all[i] = all[i << 1] + all[i
            << 1 | 1];
    }

    void update(int i, int j, int x) {
        int p = i, q = j;
        for (i += n, j += n; i < j; i >>= 1, j >>= 1) {
            if (i & 1) cnt[i] += x, recalc(i++);
            if (j & 1) cnt[--j] += x, recalc(j);
        }
        i = p + n;
        while (i >>= 1) recalc(i);
        j = q + n - 1;
        while (j >>= 1) recalc(j);
    }

    int query() {
        return curr[1];
    }
};

struct cover_tree {
    int n;
    vector<int> all, cnt, curr, xs;

    void build(int p, int l, int r) {
        all[p] = xs[r] - xs[l];
        if (l + 1 == r) return;
        int m = (l + r) / 2;
        build(2 * p, l, m);
        build(2 * p + 1, m, r);
    }

    void recalc(int p, int l, int r) {
        if (cnt[p] > 0) {
            curr[p] = all[p];
        } else if (l + 1 == r) {
            curr[p] = 0;
        } else {
            curr[p] = curr[2 * p] + curr[2 * p + 1];
        }
    }

    cover_tree(vector<int> &x): n(x.size() - 1), all(4 * n), cnt(4 *
        n), curr(4 * n), xs(x) {
        build(1, 0, n);
    }

    void update(int p, int l, int r, int ql, int qr, int val) {
        if (qr <= l or r <= ql) return;
        if (ql <= l and r <= qr) {
            cnt[p] += val;
            recalc(p, l, r);
            return;
        }
        int m = (l + r) / 2;
        update(2 * p, l, m, ql, qr, val);
        update(2 * p + 1, m, r, ql, qr, val);
        recalc(p, l, r);
    }

    int query(int p, int l, int r, int ql, int qr, int c) {
        if (qr <= l or r <= ql) return 0;
        if (ql <= l and r <= qr) {
            if (cnt[p] > 0 or c > 0) return all[p];
            return curr[p];
        }
        int m = (l + r) / 2;
        return query(2 * p, l, m, ql, qr, c + cnt[p]) + query(2 * p +
            1, m, r, ql, qr, c + cnt[p]);
    }

    void update(int l, int r, int v) {
        update(1, 0, n, l, r, v);
    }

    int query(int l, int r) {
        return query(1, 0, n, l, r, 0);
    }

    int query() {
        return curr[1];
    }
};
```

10.6 k-cover-tree

```
// Iterative K-Cover Segment Tree: calculates total length covered by
    at least K intervals
// Time: update O(K log N), query O(1) | Space: O(N * K)
struct k_cover_tree {
    int n, k;
    vector<int> all, cnt;
    vector<vector<int>> curr;

    void recalc(int i) {
        vector<int> base(k + 1);
        base[0] = all[i];
        if (i < n) for (int h = 1; h <= k; ++h) base[h] = curr[i << 1
            ][h] + curr[i << 1 | 1][h];
        for (int h = 1; h <= k; ++h) curr[i][h] = base[max<int>(0, h
            - cnt[i])];
    }

    k_cover_tree(int k, vector<int> &x): n(x.size() - 1), k(k), all(2
        * n), cnt(2 * n), curr(2 * n, vector<int>(k + 1)) {
        for (int i = 0; i < n; ++i) all[n + i] = x[i + 1] - x[i];
        for (int i = n - 1; i > 0; --i) all[i] = all[i << 1] + all[i
            << 1 | 1];
    }

    void update(int i, int j, int x) {
        int p = i, q = j;
        for (i += n, j += n; i < j; i >>= 1, j >>= 1) {
            if (i & 1) cnt[i] += x, recalc(i++);
            if (j & 1) cnt[--j] += x, recalc(j);
        }
        i = p + n;
        while (i >>= 1) recalc(i);
        j = q + n - 1;
        while (j >>= 1) recalc(j);
    }

    int query() {
        return curr[1][k];
    }
};
```

10.7 xor-cover-tree

```
// XOR Cover Tree: calculates total length covered by an odd number
    of intervals
// Time: update O(log N), query O(1) | Space: O(N)
struct xor_cover_tree {
    int n;
    vector<int> all, cnt, curr;

    void recalc(int i) {
        int child = 0;
        if (i < n) child = curr[i << 1] + curr[i << 1 | 1];
        if (cnt[i] & 1) curr[i] = all[i] - child;
        else curr[i] = child;
    }

    xor_cover_tree(vector<int> &x): n(x.size() - 1), all(2 * n),
        cnt(2 * n), curr(2 * n) {
        for (int i = 0; i < n; ++i) all[n + i] = x[i + 1] - x[i];
        for (int i = n - 1; i > 0; --i) all[i] = all[i << 1] + all[i
            << 1 | 1];
    }

    void update(int i, int j) {
        int p = i, q = j;
        for (i += n, j += n; i < j; i >>= 1, j >>= 1) {
            if (i & 1) cnt[i] ^= 1, recalc(i++);
            if (j & 1) cnt[--j] ^= 1, recalc(j);
        }
        i = p + n;
        while (i >>= 1) recalc(i);
        j = q + n - 1;
        while (j >>= 1) recalc(j);
    }

    int query() {
        return curr[1];
    }
};
```

10.8 coprime-set

```
// Coprime Set: counts elements coprime to a given number using
    Inclusion-Exclusion
// Time: insert/remove/query O(2^K * (K + log D)) | Space: O(N * 2^K)
// Note: P must be a vector of DISTINCT prime factors | K = size of
    P, D = unique divisors in map
struct coprime_set {
    map<int, int> n;

    vector<pair<int, int>> divisors(vector<int> &p) {
        int k = p.size();
        vector<pair<int, int>> a; a.reserve(1 << k);
        for (int m = 0; m < (1 << k); ++m) {
            int d = 1, mu = 1;
            for (int i = 0; i < k; ++i) if (m & (1 << i)) {
                d *= p[i], mu *= -1;
            }
            a.push_back({d, mu});
        }
        return a;
    }

    void insert(vector<int> &p) {
        vector<pair<int, int>> a = divisors(p);
        for (auto &[d, mu] : a) {
            n[d]++;
        }
    }
```

```
    void remove(vector<int> &p) {
        vector<pair<int, int>> a = divisors(p);
        for (auto &[d, mu] : a) {
            n[d]--;
        }
    }

    int query(vector<int> &p) {
        int t = 0;
        vector<pair<int, int>> a = divisors(p);
        for (auto &[d, mu] : a) {
            t += n[d] * mu;
        }
        return t;
    }
};
```

10.9 interval-set

```
// Interval Set: maintains disjoint intervals with dynamic active
     elements count
// Time: insert/remove O(log N) amortized | contains O(log N) | total
     O(1) | Space: O(N)
// Note: discrete=true for [l, r] (r-l+1) | discrete=false for
     continuous [l, r) (r-l)
struct interval_set {
    int active = 0, off;
    set<pair<int, int>> s;

    interval_set(bool discrete): off(discrete) {}

    void insert(int l, int r) {
        assert(l < r + off);
        auto itr = s.lower_bound({l, -inf});
        if (itr != s.begin() and prev(itr)->second >= l - off) itr--;
        int nl = l;
        int nr = r;
        while (itr != s.end() and itr->first <= r + off) {
            nl = min(nl, itr->first);
            nr = max(nr, itr->second);
            active -= itr->second - itr->first + off;
            itr = s.erase(itr);
        }
        active += nr - nl + off;
        s.insert({nl, nr});
    }

    void remove(int l, int r) {
        assert(l < r + off);
        auto itr = s.lower_bound({l, -inf});
        if (itr != s.begin() and prev(itr)->second >= l) itr--;
        while (itr != s.end() and itr->first <= r) {
            int cl = itr->first;
            int cr = itr->second;
            active -= cr - cl + off;
            itr = s.erase(itr);
            if (cl < l) {
                s.insert({cl, l - off});
                active += l - cl;
            }
            if (cr > r) {
                s.insert({r + off, cr});
                active += cr - r;
            }
        }
    }

    bool contains(int l, int r) {
        assert(l < r + off);
        auto itr = s.upper_bound({l, inf});
        if (itr == s.begin()) return false;
        itr--;
        return itr->first <= l and r <= itr->second;
    }

    int total() {
        return active;
    }
};
```

10.10 range-set

```
// Range Set: bitset optimized for range updates and queries
// Time: O(N/64) per operation | Space: O(N/64)
// Note: uint = uint64_t (unsigned long long)
struct range_set {
    int n;
    vector<uint> data;

    range_set(int n): n(n), data((n + 63) / 64) {}

    uint from(int i) {
        return ~0ULL << (i & 63);
    }

    uint until(int i) {
        return ~0ULL >> (63 - (i & 63));
    }

    void apply(int k, bool on, uint mask) {
        if (on) data[k] |= mask;
        else data[k] &= ~mask;
    }

    void update(int l, int r, bool on) {
        int i = l >> 6, j = r >> 6;
        if (i == j) {
            apply(i, on, from(l) & until(r));
        } else {
            apply(i, on, from(l));
            for (int k = i + 1; k < j; ++k) apply(k, on, ~0ULL);
            apply(j, on, until(r));
        }
    }

    int count(int l, int r) {
        int c = 0;
        int i = l >> 6, j = r >> 6;
        if (i == j) {
            c = __builtin_popcountll(data[i] & (from(l) & until(r)));
        } else {
            c += __builtin_popcountll(data[i] & from(l));
            for (int k = i + 1; k < j; ++k) c +=
                  __builtin_popcountll(data[k]);
            c += __builtin_popcountll(data[j] & until(r));
        }
        return c;
    }

    bool empty() {
        for (uint mask : data) if (mask) return false;
        return true;
    }

    int prev(int i) {
        for (int k = i >> 6; k >= 0; --k, i = 63) {
            uint mask = data[k] & until(i);
            if (mask) return (k << 6) + 63 - __builtin_clzll(mask);
        }
        return n;
    }

    int next(int i) {
        for (int k = i >> 6; k < data.size(); ++k, i = 0) {
            uint mask = data[k] & from(i);
            if (mask) return (k << 6) + __builtin_ctzll(mask);
        }
        return n;
    }

    int segments() {
        int s = 0;
        for (int k = 0; k < data.size(); ++k) {
            s += __builtin_popcountll((data[k] >> 1) & ~data[k]);
            if ((data[k] & 1) and (k == 0 or (data[k - 1] >> 63) == 0
                 )) s += 1;
        }
        return s;
    }
};
```

10.11 venice-set

```
// Venice Set: multiset with global value updates
// Time: O(log N) | Space: O(N)
struct venice_set {
    int k = 0;
    multiset<int> s;

    void insert(int x) {
        s.insert(x - k);
    }

    void remove(int x) {
        if (auto i = s.find(x - k); i != s.end()) s.erase(i);
    }

    void update(int x) {
        k += x;
    }

    int min() {
        return *s.begin() + k;
    }
};
```

10.12 disjoint-set-union-2d

```
struct dsu_2d {
    int n, m;
    dsu d;

    dsu_2d(int n, int m): n(n), m(m), d(n * m) {}

    int id(int r, int c) {
        return r * m + c;
    }

    int find(int r, int c) {
        return d.find(id(r, c));
    }

    void unite(int r1, int c1, int r2, int c2) {
        d.unite(id(r1, c1), id(r2, c2));
    }
};
```

10.13 disjoint-set-union

```
struct dsu {
    vector<int> p, s;

    dsu(int n): p(n), s(n, 1) {
        iota(p.begin(), p.end(), 0);
    }

    int find(int a) {
        if (p[a] == a) return a;
        return p[a] = find(p[a]);
    }
```

```
    void unite(int a, int b) {
        a = find(a);
        b = find(b);
        if (a == b) return;
        if (s[a] < s[b]) swap(a, b);
        p[b] = a;
        s[a] += s[b];
    }
};
```

10.14 dynamic-connectivity

```
struct dynamic_connectivity {
private:
    int n, q;
    vector<vector<pair<int, int>>> tree;
    rollback_dsu dsu;

    void insert(pair<int, int> e, int ql, int qr, int p, int l, int
         r) {
        if (qr < l or ql > r) return;
        if (ql <= l and qr >= r) return void(tree[p].push_back(e));
        int m = (l + r) / 2;
        insert(e, ql, qr, 2 * p, l, m);
        insert(e, ql, qr, 2 * p + 1, m + 1, r);
    }

    void compute(vector<int> &ans, int p, int l, int r) {
        for (auto [a, b] : tree[p]) dsu.unite(a, b);
        if (l == r) {
            ans[l] = dsu.c;
        } else {
            int m = (l + r) / 2;
            compute(ans, 2 * p, l, m);
            compute(ans, 2 * p + 1, m + 1, r);
        }
        for (auto [a, b] : tree[p]) dsu.rollback();
    }

public:
    dynamic_connectivity(int n, int q): n(n), q(q), tree(4 * q),
         dsu(n) {}

    void insert(pair<int, int> e, int ql, int qr) {
        insert(e, ql, qr, 1, 0, q - 1);
    }

    vector<int> compute() {
        vector<int> ans(q);
        compute(ans, 1, 0, q - 1);
        return ans;
    }
};
```

10.15 rollback-disjoint-set-union

```
struct rollback_dsu {
    int c;
    vector<int> p, s;
    vector<pair<int, int>> h;

    rollback_dsu(int n): c(n), p(n), s(n, 1) {
        iota(p.begin(), p.end(), 0);
    }

    int find(int a) {
        while (p[a] != a) a = p[a];
        return a;
    }

    void unite(int a, int b) {
        a = find(a);
        b = find(b);
        if (a == b) {
            h.push_back({-1, -1});
        } else {
            if (s[a] < s[b]) swap(a, b);
            h.push_back({a, b});
            c--;
            p[b] = a;
            s[a] += s[b];
        }
    }

    void rollback() {
        assert(h.size());
        auto [a, b] = h.back(); h.pop_back();
        if (a != -1) {
            c++;
            p[b] = b;
            s[a] -= s[b];
        }
    }

    int state() {
        return h.size();
    }
};
```

10.16 dynamic-merge-sort-tree

```
// Merge Sort Tree: range queries with point updates
// Time: build O(N log^2 N) | Update/Query: O(log^2 N) | Space: O(N
     log N)
// Note: uses PBDS ordered_set | pair<int, int> handles duplicate
     values
struct merge_sort_tree {
    int n;
    vector<int> a;
    vector<ordered_set<pair<int, int>>> tree;

    merge_sort_tree(vector<int> &a): n(a.size()), a(a), tree(2 * n) {
        for (int i = 0; i < n; ++i) {
            for (int j = i + n; j > 0; j >>= 1) {
                tree[j].insert({a[i], i});
            }
        }
    }

    void update(int i, int x) {
        for (int j = i + n; j > 0; j >>= 1) {
            tree[j].erase({a[i], i});
            tree[j].insert({x, i});
        }
        a[i] = x;
    }

    int less_or_equal(int i, int j, int x) {
        int c = 0;
        for (i += n, j += n + 1; i < j; i >>= 1, j >>= 1) {
            if (i & 1) c += tree[i++].order_of_key({x + 1, 0});
            if (j & 1) c += tree[--j].order_of_key({x + 1, 0});
        }
        return c;
    }
};
```

10.17 merge-sort-tree

```
struct merge_sort_tree {
    int n;
    vector<vector<int>> tree;

    merge_sort_tree(vector<int> &a): n(a.size()), tree(2 * n) {
        for (int i = 0; i < n; ++i) tree[i + n] = {a[i]};
        for (int i = n - 1; i > 0; --i) merge(tree[i << 1].begin(),
             tree[i << 1].end(), tree[(i << 1) | 1].begin(), tree[(i
             << 1) | 1].end(), back_inserter(tree[i]));
    }

    int less_or_equal(int i, int j, int x) {
        int c = 0;
        for (i += n, j += n + 1; i < j; i >>= 1, j >>= 1) {
            if (i & 1) c += upper_bound(tree[i].begin(),
                 tree[i].end(), x) - tree[i].begin(), i++;
            if (j & 1) --j, c += upper_bound(tree[j].begin(),
                 tree[j].end(), x) - tree[j].begin();
        }
        return c;
    }
};
```

10.18 hash-table

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;

struct custom_hash {
    size_t operator()(size_t x) const {
        static const size_t time =
             chrono::steady_clock::now().time_since_epoch().count();
        return (x + 0x9e3779b9) ^ time;
    }
};

template<typename T>
using hash_set = gp_hash_table<T, null_type, custom_hash>;

template<typename K, typename V>
using hash_map = gp_hash_table<K, V, custom_hash>;
```

10.19 ordered-set

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;

template <typename T>
using ordered_set = tree<T, null_type, less<T>, rb_tree_tag,
     tree_order_statistics_node_update>;
```

10.20 lazy-segment-tree

```
struct lazy_segment_tree {
private:
    int n;
    vector<int> tree, lazy;

    void apply(int p, int x, int l) {
        tree[p] += x * l;
        lazy[p] += x;
    }

    void push(int p, int l, int r) {
        if (lazy[p] == 0) return;
        int m = (l + r) / 2;
        apply(2 * p, lazy[p], m - l + 1);
        apply(2 * p + 1, lazy[p], r - m);
        lazy[p] = 0;
    }

    void pull(int p) {
        tree[p] = tree[2 * p] + tree[2 * p + 1];
    }

    void build(int p, int l, int r, vector<int> &a) {
        if (l == r) {
            tree[p] = a[l];
            return;
        }
        int m = (l + r) / 2;
        build(2 * p, l, m, a);
```

```
        build(2 * p + 1, m + 1, r, a);
        pull(p);
    }

    void update(int p, int l, int r, int ql, int qr, int x) {
        if (qr < l or r < ql) return;
        if (ql <= l and r <= qr) {
            apply(p, x, r - l + 1);
            return;
        }
        push(p, l, r);
        int m = (l + r) / 2;
        update(2 * p, l, m, ql, qr, x);
        update(2 * p + 1, m + 1, r, ql, qr, x);
        pull(p);
    }

    int query(int p, int l, int r, int ql, int qr) {
        if (qr < l or r < ql) return 0;
        if (ql <= l and r <= qr) return tree[p];
        push(p, l, r);
        int m = (l + r) / 2;
        return query(2 * p, l, m, ql, qr) + query(2 * p + 1, m + 1,
             r, ql, qr);
    }

public:
    lazy_segment_tree(int n): n(n), tree(4 * n), lazy(4 * n) {}

    lazy_segment_tree(vector<int> &a): lazy_segment_tree(a.size()) {
        build(1, 0, n - 1, a);
    }

    void update(int l, int r, int x) {
        update(1, 0, n - 1, l, r, x);
    }

    int query(int l, int r) {
        return query(1, 0, n - 1, l, r);
    }
};
```

10.21 lazy-sparse-segment-tree

```
struct lazy_sparse_segment_tree {
private:
    struct node {
        int val = 0, lazy = 0;
        int l = 0, r = 0;
    };

    int root = 0, min_l, max_r;
    vector<node> tree;

    void push(int p, int l, int r) {
        if (not tree[p].lazy) return;
        int m = l + (r - l) / 2;
        if (not tree[p].l) {
            tree[p].l = tree.size();
            tree.emplace_back();
        }
        if (not tree[p].r) {
            tree[p].r = tree.size();
            tree.emplace_back();
        }
        tree[tree[p].l].lazy += tree[p].lazy;
        tree[tree[p].r].lazy += tree[p].lazy;
        tree[tree[p].l].val += tree[p].lazy * (m - l + 1);
        tree[tree[p].r].val += tree[p].lazy * (r - m);
        tree[p].lazy = 0;
    }

    int update(int ql, int qr, int x, int p, int l, int r) {
        if (not p) {
            p = tree.size();
            tree.emplace_back();
        }
        if (ql <= l and r <= qr) {
            tree[p].lazy += x;
            tree[p].val += x * (r - l + 1);
            return p;
        }
        push(p, l, r);
        int m = l + (r - l) / 2;
        if (ql <= m) {
            int nl = update(ql, qr, x, tree[p].l, l, m);
            tree[p].l = nl;
        }
        if (qr > m) {
            int nr = update(ql, qr, x, tree[p].r, m + 1, r);
            tree[p].r = nr;
        }
        tree[p].val = tree[tree[p].l].val + tree[tree[p].r].val;
        return p;
    }

    int query(int ql, int qr, int p, int l, int r) {
        if (not p or (qr < l or r < ql)) {
            return 0;
        }
        if (ql <= l and r <= qr) {
            return tree[p].val;
        }
        push(p, l, r);
        int m = l + (r - l) / 2;
        return query(ql, qr, tree[p].l, l, m) + query(ql, qr,
             tree[p].r, m + 1, r);
    }

public:
    lazy_sparse_segment_tree(int l, int r, int q = 0): min_l(l),
         max_r(r) {
        tree.reserve(1 + 35 * q);
        tree.emplace_back();
    }

    void update(int ql, int qr, int x) {
        root = update(ql, qr, x, root, min_l, max_r);
    }

    int query(int ql, int qr) {
        return query(ql, qr, root, min_l, max_r);
    }
};
```

10.22 persistent-segment-tree

```
struct persistent_segment_tree {
private:
    struct node {
        int x = 0;
        int l = 0, r = 0;
    };

    int n;
    vector<node> tree;

    int clone(int p) {
        tree.push_back(tree[p]);
        return tree.size() - 1;
    }

    int build(vector<int> &a, int l, int r) {
        int p = tree.size();
        tree.emplace_back();
        if (l == r) {
            tree[p].x = a[l];
            return p;
        }
        int m = (l + r) / 2;
        tree[p].l = build(a, l, m);
        tree[p].r = build(a, m + 1, r);
        tree[p].x = tree[tree[p].l].x + tree[tree[p].r].x;
        return p;
    }

    int update(int i, int x, int p, int l, int r) {
        int c = clone(p);
        if (l == r) {
            tree[c].x = x;
            return c;
        }
        int m = (l + r) / 2;
        if (i <= m) {
            int nl = update(i, x, tree[c].l, l, m);
            tree[c].l = nl;
        } else {
            int nr = update(i, x, tree[c].r, m + 1, r);
            tree[c].r = nr;
        }
        tree[c].x = tree[tree[c].l].x + tree[tree[c].r].x;
        return c;
    }

    int query(int ql, int qr, int p, int l, int r) {
        if (qr < l or r < ql) return 0;
        if (ql <= l and r <= qr) return tree[p].x;
        int m = (l + r) / 2;
        return query(ql, qr, tree[p].l, l, m) + query(ql, qr,
             tree[p].r, m + 1, r);
    }

    int copy_range(int ql, int qr, int s, int t, int l, int r) {
        if (qr < l or r < ql) return t;
        if (ql <= l and r <= qr) return s;
        int c = clone(t);
        int m = (l + r) / 2;
        int nl = copy_range(ql, qr, tree[s].l, tree[t].l, l, m);
        tree[c].l = nl;
        int nr = copy_range(ql, qr, tree[s].r, tree[t].r, m + 1, r);
        tree[c].r = nr;
        tree[c].x = tree[tree[c].l].x + tree[tree[c].r].x;
        return c;
    }

public:
    vector<int> roots;

    persistent_segment_tree(vector<int> &a, int q = 0): n(a.size()) {
        tree.reserve(4 * n + 20 * q);
        roots.push_back(build(a, 0, n - 1));
    }

    void update(int i, int x, int v = -1) {
        if (v == -1) v = roots.size() - 1;
        roots.push_back(update(i, x, roots[v], 0, n - 1));
    }

    int query(int ql, int qr, int v = -1) {
        if (v == -1) v = roots.size() - 1;
        return query(ql, qr, roots[v], 0, n - 1);
    }

    void copy_range(int ql, int qr, int s, int t = -1) {
        if (t == -1) t = roots.size() - 1;
        roots.push_back(copy_range(ql, qr, roots[s], roots[t], 0, n -
             1));
    }
};
```

10.23 segment-tree-2d

```
struct segment_tree_2d {
    int n, m;
    vector<segment_tree> tree;

    segment_tree_2d(int n, int m): n(n), m(m), tree(2 * n,
         segment_tree(m)) {}

    segment_tree_2d(vector<vector<int>> &a):
         segment_tree_2d(a.size(), a[0].size()) {
        for (int i = 0; i < n; ++i) tree[n + i] = segment_tree(a[i]);
        for (int i = n - 1; i > 0; --i) {
            for (int j = 1; j < 2 * m; ++j) {
                tree[i].tree[j] = segment_tree::merge(tree[i << 1
                     ].tree[j], tree[(i << 1) | 1].tree[j]);
            }
        }
    }

    void update(int i, int j, int x) {
        tree[i += n].update(j, x);
        while (i >>= 1) tree[i].update(j, segment_tree::merge(tree[i
             << 1].tree[j + m], tree[(i << 1) | 1].tree[j + m]));
    }

    int query(int i1, int j1, int i2, int j2) {
        int l = 0, r = 0;
        for (i1 += n, i2 += n + 1; i1 < i2; i1 >>= 1, i2 >>= 1) {
            if (i1 & 1) l = segment_tree::merge(l,
                 tree[i1++].query(j1, j2));
            if (i2 & 1) r = segment_tree::merge(tree[--i2].query(j1,
                 j2), r);
        }
        return segment_tree::merge(l, r);
    }
};
```

10.24 segment-tree-beats

```
// Segment Tree Beats: advanced range updates via condition pruning
// Time: update amortized O(log^2 N), query O(log N) | Space: O(N)
// Note: supports sum, add, chmin, chmax and mod over a range
struct segment_tree_beats {
private:
    struct node {
        int sum, lazy;
        int max1, max2, maxc;
        int min1, min2, minc;
    };

    int n;
    vector<node> tree;

    void push_up(int p) {
        int l = 2 * p, r = 2 * p + 1;
        tree[p].sum = tree[l].sum + tree[r].sum;

        if (tree[l].max1 == tree[r].max1) {
            tree[p].max1 = tree[l].max1;
            tree[p].max2 = max(tree[l].max2, tree[r].max2);
            tree[p].maxc = tree[l].maxc + tree[r].maxc;
        } else if (tree[l].max1 > tree[r].max1) {
            tree[p].max1 = tree[l].max1;
            tree[p].max2 = max(tree[l].max2, tree[r].max1);
            tree[p].maxc = tree[l].maxc;
        } else {
            tree[p].max1 = tree[r].max1;
            tree[p].max2 = max(tree[l].max1, tree[r].max2);
            tree[p].maxc = tree[r].maxc;
        }

        if (tree[l].min1 == tree[r].min1) {
            tree[p].min1 = tree[l].min1;
            tree[p].min2 = min(tree[l].min2, tree[r].min2);
            tree[p].minc = tree[l].minc + tree[r].minc;
        } else if (tree[l].min1 < tree[r].min1) {
            tree[p].min1 = tree[l].min1;
            tree[p].min2 = min(tree[l].min2, tree[r].min1);
            tree[p].minc = tree[l].minc;
        } else {
            tree[p].min1 = tree[r].min1;
            tree[p].min2 = min(tree[l].min1, tree[r].min2);
            tree[p].minc = tree[r].minc;
        }
    }

    void apply_add(int p, int l, int r, int v) {
        tree[p].sum += v * (r - l + 1);
        tree[p].max1 += v;
        if (tree[p].max2 != -inf) tree[p].max2 += v;
        tree[p].min1 += v;
        if (tree[p].min2 != inf) tree[p].min2 += v;
        tree[p].lazy += v;
    }

    void apply_min(int p, int v) {
        if (v >= tree[p].max1) return;
        tree[p].sum -= (tree[p].max1 - v) * tree[p].maxc;
        if (tree[p].min1 == tree[p].max1) tree[p].min1 = v;
        if (tree[p].min2 == tree[p].max1) tree[p].min2 = v;
        tree[p].max1 = v;
    }

    void apply_max(int p, int v) {
        if (v <= tree[p].min1) return;
        tree[p].sum += (v - tree[p].min1) * tree[p].minc;
        if (tree[p].max1 == tree[p].min1) tree[p].max1 = v;
        if (tree[p].max2 == tree[p].min1) tree[p].max2 = v;
        tree[p].min1 = v;
    }

    void push_down(int p, int l, int r) {
        int m = (l + r) / 2;
        int left = 2 * p, right = 2 * p + 1;

        if (tree[p].lazy != 0) {
            apply_add(left, l, m, tree[p].lazy);
            apply_add(right, m + 1, r, tree[p].lazy);
            tree[p].lazy = 0;
        }

        apply_min(left, tree[p].max1);
        apply_min(right, tree[p].max1);

        apply_max(left, tree[p].min1);
        apply_max(right, tree[p].min1);
    }

    void build(const vector<int> &a, int p, int l, int r) {
        tree[p].lazy = 0;
        if (l == r) {
            tree[p].sum = tree[p].max1 = tree[p].min1 = a[l];
            tree[p].maxc = tree[p].minc = 1;
            tree[p].max2 = -inf;
            tree[p].min2 = inf;
        } else {
            int m = (l + r) / 2;
            build(a, 2 * p, l, m);
            build(a, 2 * p + 1, m + 1, r);
            push_up(p);
        }
    }

    void update_add(int ql, int qr, int v, int p, int l, int r) {
        if (ql <= l and r <= qr) {
            apply_add(p, l, r, v);
            return;
        }
        push_down(p, l, r);
        int m = (l + r) / 2;
        if (ql <= m) update_add(ql, qr, v, 2 * p, l, m);
        if (qr > m) update_add(ql, qr, v, 2 * p + 1, m + 1, r);
        push_up(p);
    }

    void update_min(int ql, int qr, int v, int p, int l, int r) {
        if (v >= tree[p].max1) return;
        if (ql <= l and r <= qr and v > tree[p].max2) {
            apply_min(p, v);
            return;
        }
        push_down(p, l, r);
        int m = (l + r) / 2;
        if (ql <= m) update_min(ql, qr, v, 2 * p, l, m);
        if (qr > m) update_min(ql, qr, v, 2 * p + 1, m + 1, r);
        push_up(p);
    }

    void update_max(int ql, int qr, int v, int p, int l, int r) {
        if (v <= tree[p].min1) return;
        if (ql <= l and r <= qr and v < tree[p].min2) {
            apply_max(p, v);
            return;
        }
        push_down(p, l, r);
        int m = (l + r) / 2;
        if (ql <= m) update_max(ql, qr, v, 2 * p, l, m);
        if (qr > m) update_max(ql, qr, v, 2 * p + 1, m + 1, r);
        push_up(p);
    }

    void update_mod(int ql, int qr, int v, int p, int l, int r) {
        if (ql <= l and r <= qr) {
            if (tree[p].max1 < v) return;
            if (tree[p].min1 == tree[p].max1) {
                apply_add(p, l, r, (tree[p].max1 % v) - tree[p].max1);
                return;
            }
        }
        push_down(p, l, r);
        int m = (l + r) / 2;
        if (ql <= m) update_mod(ql, qr, v, 2 * p, l, m);
        if (qr > m) update_mod(ql, qr, v, 2 * p + 1, m + 1, r);
        push_up(p);
    }

    int query_sum(int ql, int qr, int p, int l, int r) {
        if (ql <= l and r <= qr) return tree[p].sum;
        push_down(p, l, r);
        int m = (l + r) / 2;
        int ans = 0;
        if (ql <= m) ans += query_sum(ql, qr, 2 * p, l, m);
        if (qr > m) ans += query_sum(ql, qr, 2 * p + 1, m + 1, r);
        return ans;
    }

public:
    segment_tree_beats(int n) : n(n), tree(4 * n) {}

    segment_tree_beats(const vector<int> &a) : n(a.size()), tree(4 *
         n) {
        build(a, 1, 0, n - 1);
    }

    void update_add(int l, int r, int v) {
        update_add(l, r, v, 1, 0, n - 1);
    }
```

```
    void update_min(int l, int r, int v) {
        update_min(l, r, v, 1, 0, n - 1);
    }

    void update_max(int l, int r, int v) {
        update_max(l, r, v, 1, 0, n - 1);
    }

    void update_mod(int l, int r, int v) {
        update_mod(l, r, v, 1, 0, n - 1);
    }

    int query_sum(int l, int r) {
        return query_sum(l, r, 1, 0, n - 1);
    }
};
```

10.25 segment-tree

```
struct segment_tree {
    int n;
    vector<int> tree;

    int merge(int x, int y) {
        return x + y;
    }

    segment_tree(int s): n(__bit_ceil(s)), tree(2 * n) {}

    segment_tree(vector<int> &a): segment_tree(a.size()) {
        for (int i = 0; i < a.size(); ++i) tree[n + i] = a[i];
        for (int i = n - 1; i > 0; --i) tree[i] = merge(tree[i << 1],
             tree[(i << 1) | 1]);
    }

    void update(int i, int x) {
        i += n;
        tree[i] = x;
        while (i >>= 1) tree[i] = merge(tree[i << 1], tree[(i << 1) |
             1]);
    }

    int query(int i, int j) {
        int l = 0, r = 0;
        for (i += n, j += n + 1; i < j; i >>= 1, j >>= 1) {
            if (i & 1) l = merge(l, tree[i++]);
            if (j & 1) r = merge(tree[--j], r);
        }
        return merge(l, r);
    }
};

struct segment_tree {
private:
    int n;
    vector<int> tree;

    int merge(int x, int y) {
        return x + y;
    }

    void build(int p, int l, int r, vector<int> &a) {
        if (l == r) {
            tree[p] = a[l];
            return;
        }
        int m = (l + r) / 2;
        build(2 * p, l, m, a);
        build(2 * p + 1, m + 1, r, a);
        tree[p] = merge(tree[2 * p], tree[2 * p + 1]);
    }

    void update(int p, int l, int r, int i, int x) {
        if (i < l or r < i) return;
        if (l == r) {
            tree[p] = x;
            return;
        }
        int m = (l + r) / 2;
        update(2 * p, l, m, i, x);
        update(2 * p + 1, m + 1, r, i, x);
        tree[p] = merge(tree[2 * p], tree[2 * p + 1]);
    }

    int query(int p, int l, int r, int ql, int qr) {
        if (qr < l or r < ql) return 0;
        if (ql <= l and r <= qr) return tree[p];
        int m = (l + r) / 2;
        return merge(query(2 * p, l, m, ql, qr), query(2 * p + 1, m +
             1, r, ql, qr));
    }

public:
    segment_tree(int n): n(n), tree(4 * n) {}

    segment_tree(vector<int> &a): n(a.size()), tree(4 * n) {
        build(1, 0, n - 1, a);
    }

    void update(int i, int x) {
        update(1, 0, n - 1, i, x);
    }

    int query(int l, int r) {
        return query(1, 0, n - 1, l, r);
    }
};
```

10.26 sparse-persistent-segment-tree

```
struct sparse_persistent_segment_tree {
private:
    struct node {
        int x = 0;
        int l = 0, r = 0;
    };

    int min_l, max_r;
    vector<node> tree;

    int clone(int p) {
        tree.push_back(tree[p]);
        return tree.size() - 1;
    }

    int update(int i, int x, int p, int l, int r) {
        int c = clone(p);
        if (l == r) {
            tree[c].x = x;
            return c;
        }
        int m = l + (r - l) / 2;
        if (i <= m) {
            int nl = update(i, x, tree[c].l, l, m);
            tree[c].l = nl;
        } else {
            int nr = update(i, x, tree[c].r, m + 1, r);
            tree[c].r = nr;
        }
        tree[c].x = tree[tree[c].l].x + tree[tree[c].r].x;
        return c;
    }

    int query(int ql, int qr, int p, int l, int r) {
        if (not p or (qr < l or r < ql)) return 0;
        if (ql <= l and r <= qr) return tree[p].x;
        int m = l + (r - l) / 2;
        return query(ql, qr, tree[p].l, l, m) + query(ql, qr,
             tree[p].r, m + 1, r);
    }

public:
    vector<int> roots;

    sparse_persistent_segment_tree(int l, int r, int q = 0):
         min_l(l), max_r(r) {
        tree.reserve(1 + 35 * q);
        tree.emplace_back();
        roots.push_back(0);
    }

    void update(int i, int x, int v = -1) {
        if (v == -1) v = roots.size() - 1;
        roots.push_back(update(i, x, roots[v], min_l, max_r));
    }

    int query(int ql, int qr, int v = -1) {
        if (v == -1) v = roots.size() - 1;
        return query(ql, qr, roots[v], min_l, max_r);
    }
};
```

10.27 sparse-segment-tree

```
struct sparse_segment_tree {
private:
    struct node {
        int x = 0;
        int l = 0, r = 0;
    };

    int root = 0, min_l, max_r;
    vector<node> tree;

    int update(int i, int x, int p, int l, int r) {
        if (not p) {
            p = tree.size();
            tree.emplace_back();
        }
        if (l == r) {
            tree[p].x = x;
            return p;
        }
        int m = l + (r - l) / 2;
        if (i <= m) {
            int nl = update(i, x, tree[p].l, l, m);
            tree[p].l = nl;
        } else {
            int nr = update(i, x, tree[p].r, m + 1, r);
            tree[p].r = nr;
        }

        tree[p].x = tree[tree[p].l].x + tree[tree[p].r].x;
        return p;
    }

    int query(int ql, int qr, int p, int l, int r) {
        if (not p or (qr < l or r < ql)) {
            return 0;
        }
        if (ql <= l and r <= qr) {
            return tree[p].x;
        }
        int m = l + (r - l) / 2;
        return query(ql, qr, tree[p].l, l, m) + query(ql, qr,
             tree[p].r, m + 1, r);
    }

public:
```

```
    sparse_segment_tree(int l, int r, int q = 0): min_l(l), max_r(r) {
        tree.reserve(1 + 35 * q);
        tree.emplace_back();
    }

    void update(int i, int x) {
        root = update(i, x, root, min_l, max_r);
    }

    int query(int ql, int qr) {
        return query(ql, qr, root, min_l, max_r);
    }
};
```

10.28 sparse-table-2d

```
struct sparse_table_2d {
    int n, m, l;
    vector<vector<sparse_table>> table;

    sparse_table_2d(vector<vector<int>> &a): n(a.size()),
         m(a[0].size()), l(__lg(n)), table(l + 1) {
        for (int i = 0; i < n; ++i) {
            table[0].push_back(sparse_table(a[i]));
        }
        for (int k = 1; k <= l; ++k) {
            for (int i = 0; i + (1 << k) <= n; ++i) {
                vector<int> h(m);
                for (int j = 0; j < m; ++j) {
                    int x = table[k - 1][i].table[0][j];
                    int y = table[k - 1][i + (1 << (k - 1
                         ))].table[0][j];
                    h[j] = sparse_table::merge(x, y);
                }
                table[k].push_back(sparse_table(h));
            }
        }
    }

    int query(int i1, int j1, int i2, int j2) {
        int k = __lg(i2 - i1 + 1);
        int x = table[k][i1].query(j1, j2);
        int y = table[k][i2 - (1 << k) + 1].query(j1, j2);
        return sparse_table::merge(x, y);
    }
};
```

10.29 sparse-table

```
struct sparse_table {
    int n, l;
    vector<vector<int>> table;

    static int merge(int x, int y) {
        return gcd(x, y);
    }

    sparse_table(vector<int> &a): n(a.size()), l(__lg(n)), table(l + 1
         , vector<int>(n)) {
        for (int i = 0; i < n; ++i) table[0][i] = a[i];
        for (int k = 1; k <= l; ++k) {
            for (int i = 0; i + (1 << k) <= n; ++i) {
                table[k][i] = merge(table[k - 1][i], table[k - 1][i +
                     (1 << (k - 1))]);
            }
        }
    }

    int query(int i, int j) {
        int k = __lg(j - i + 1);
        return merge(table[k][i], table[k][j - (1 << k) + 1]);
    }
};
```

10.30 lazy-splay-tree

```
struct lazy_node {
    int val;
    int sum;
    int lazy;
    bool rev;
    int size;
    int p;
    array<int, 2> c;

    void init(int v) {
        val = v;
        sum = v;
        lazy = 0;
        rev = false;
        size = 1;
        p = 0;
        c.fill(0);
    }
};

struct lazy_splay_tree {
    int root;
    vector<lazy_node> t;

    bool is_root(int n) {
        if (n == 0) return true;
        int p = t[n].p;
        return t[p].c[0] != n and t[p].c[1] != n;
    }

    void pull(int n) {
        if (n == 0) return;
        int l = t[n].c[0], r = t[n].c[1];
        t[n].size = 1 + t[l].size + t[r].size;
        t[n].sum = t[n].val + t[l].sum + t[r].sum;
    }

    void flip(int n) {
        if (n == 0) return;
        swap(t[n].c[0], t[n].c[1]);
        t[n].rev ^= 1;
    }

    void apply(int n, int v) {
        if (n == 0) return;
        t[n].val += v;
        t[n].sum += v * t[n].size;
        t[n].lazy += v;
    }

    void push(int n) {
        if (n == 0) return;
        if (t[n].rev == 1) {
            flip(t[n].c[0]);
            flip(t[n].c[1]);
            t[n].rev = 0;
        }
        if (t[n].lazy != 0) {
            apply(t[n].c[0], t[n].lazy);
            apply(t[n].c[1], t[n].lazy);
            t[n].lazy = 0;
        }
    }

    void push_path(int n) {
        if (not is_root(n)) push_path(t[n].p);
        push(n);
    }

    void rotate(int n) {
        int p = t[n].p;
        int g = t[p].p;
        bool d = n == t[p].c[1];
        if (not is_root(p)) t[g].c[p == t[g].c[1]] = n;
        t[n].p = g;
        t[p].c[d] = t[n].c[d ^ 1];
        if (t[n].c[d ^ 1]) t[t[n].c[d ^ 1]].p = p;
        t[n].c[d ^ 1] = p;
        t[p].p = n;
        pull(p);
        pull(n);
    }

    void splay(int n, int s) {
        if (n == 0) return;
        push_path(n);
        while (t[n].p != s and not (s == 0 and is_root(n))) {
            int p = t[n].p;
            int g = t[p].p;
            if (g != s and not (s == 0 and is_root(p))) {
                if ((t[g].c[1] == p) ^ (t[p].c[1] == n)) rotate(n);
                else rotate(p);
            }
            rotate(n);
        }
        if (s == 0 and t[n].p == 0) root = n;
    }

    int kth(int k) {
        int n = root;
        while (n != 0) {
            push(n);
            int l = t[n].c[0];
            if (k < t[l].size) {
                n = l;
            } else if (k == t[l].size) {
                return n;
            } else {
                k -= t[l].size + 1;
                n = t[n].c[1];
            }
        }
        return 0;
    }

    int merge(int l, int r) {
        if (l == 0 or r == 0) return l | r;
        root = l;
        int m = kth(t[l].size - 1);
        splay(m, 0);
        t[m].c[1] = r;
        t[r].p = m;
        pull(m);
        return m;
    }

    pair<int, int> split(int n, int k) {
        if (k <= 0) return {0, n};
        if (k >= t[n].size) return {n, 0};
        root = n;
        int r = kth(k);
        splay(r, 0);
        int l = t[r].c[0];
        if (l != 0) t[l].p = 0;
        t[r].c[0] = 0;
        pull(r);
        return {l, r};
    }

    void init() {
        root = 0;
        t.clear();
        t.emplace_back();
        t[0].init(0);
```

```
        t[0].size = 0;
    }

    void insert(int i, int v) {
        int n = t.size();
        t.emplace_back();
        t[n].init(v);
        auto [l, r] = split(root, i);
        root = merge(merge(l, n), r);
    }

    void reverse(int i, int j) {
        auto [h, r] = split(root, j + 1);
        auto [l, m] = split(h, i);
        flip(m);
        root = merge(merge(l, m), r);
    }

    void update(int i, int j, int v) {
        auto [h, r] = split(root, j + 1);
        auto [l, m] = split(h, i);
        apply(m, v);
        root = merge(merge(l, m), r);
    }

    int query(int i, int j) {
        auto [h, r] = split(root, j + 1);
        auto [l, m] = split(h, i);
        int q = t[m].sum;
        root = merge(merge(l, m), r);
        return q;
    }
};
```

10.31 splay-tree

```
struct node {
    int val;
    int sum;
    bool rev;
    int size;
    int p;
    array<int, 2> c;

    void init(int v) {
        val = v;
        sum = v;
        rev = false;
        size = 1;
        p = 0;
        c.fill(0);
    }
};

struct splay_tree {
    int root;
    vector<node> t;

    bool is_root(int n) {
        if (n == 0) return true;
        int p = t[n].p;
        return t[p].c[0] != n and t[p].c[1] != n;
    }

    void pull(int n) {
        if (n == 0) return;
        int l = t[n].c[0], r = t[n].c[1];
        t[n].size = 1 + t[l].size + t[r].size;
        t[n].sum = t[n].val + t[l].sum + t[r].sum;
    }

    void flip(int n) {
        if (n == 0) return;
        swap(t[n].c[0], t[n].c[1]);
        t[n].rev ^= 1;
    }

    void push(int n) {
        if (n == 0) return;
        if (t[n].rev == 1) {
            flip(t[n].c[0]);
            flip(t[n].c[1]);
            t[n].rev = 0;
        }
    }

    void push_path(int n) {
        if (not is_root(n)) push_path(t[n].p);
        push(n);
    }

    void rotate(int n) {
        int p = t[n].p;
        int g = t[p].p;
        bool d = n == t[p].c[1];
        if (not is_root(p)) t[g].c[p == t[g].c[1]] = n;
        t[n].p = g;
        t[p].c[d] = t[n].c[d ^ 1];
        if (t[n].c[d ^ 1]) t[t[n].c[d ^ 1]].p = p;
        t[n].c[d ^ 1] = p;
        t[p].p = n;
        pull(p);
        pull(n);
    }

    void splay(int n, int s) {
        if (n == 0) return;
        push_path(n);
        while (t[n].p != s and not (s == 0 and is_root(n))) {
            int p = t[n].p;
            int g = t[p].p;
            if (g != s and not (s == 0 and is_root(p))) {
                if ((t[g].c[1] == p) ^ (t[p].c[1] == n)) rotate(n);
                else rotate(p);
            }
            rotate(n);
        }
        if (s == 0 and t[n].p == 0) root = n;
    }

    int kth(int k) {
        int n = root;
        while (n != 0) {
            push(n);
            int l = t[n].c[0];
            if (k < t[l].size) {
                n = l;
            } else if (k == t[l].size) {
                return n;
            } else {
                k -= t[l].size + 1;
                n = t[n].c[1];
            }
        }
        return 0;
    }

    int merge(int l, int r) {
        if (l == 0 or r == 0) return l | r;
        root = l;
        int m = kth(t[l].size - 1);
        splay(m, 0);
        t[m].c[1] = r;
        t[r].p = m;
        pull(m);
        return m;
    }

    pair<int, int> split(int n, int k) {
        if (k <= 0) return {0, n};
        if (k >= t[n].size) return {n, 0};
        root = n;
        int r = kth(k);
        splay(r, 0);
        int l = t[r].c[0];
        if (l != 0) t[l].p = 0;
        t[r].c[0] = 0;
        pull(r);
        return {l, r};
    }

    void init() {
        root = 0;
        t.clear();
        t.emplace_back();
        t[0].init(0);
        t[0].size = 0;
    }

    void insert(int i, int v) {
        int n = t.size();
        t.emplace_back();
        t[n].init(v);
        auto [l, r] = split(root, i);
        root = merge(merge(l, n), r);
    }

    void reverse(int i, int j) {
        auto [h, r] = split(root, j + 1);
        auto [l, m] = split(h, i);
        flip(m);
        root = merge(merge(l, m), r);
    }

    void update(int i, int v) {
        int n = kth(i);
        splay(n, 0);
        t[n].val = v;
        pull(n);
    }

    int query(int i, int j) {
        auto [h, r] = split(root, j + 1);
        auto [l, m] = split(h, i);
        int q = t[m].sum;
        root = merge(merge(l, m), r);
        return q;
    }
};
```

10.32 implicit-treap

```
// Implicit Treap: dynamic array with range updates and queries
// Time: operations O(log N) average | Space: O(N)
// Note: supports range modifications | uses lazy propagation and
     implicit indices
struct treap {
    struct node {
        int x, p, l = 0, r = 0;
        int sum, size = 1, lazy = 0, rev = 0;

        node(int x, int p = rng()): x(x), p(p), sum(x) {}
    };

    int root = 0;
    vector<node> t;

    treap(int n = 0) {
        t.reserve(n + 1);
```

```
    t.push_back(node(0));
    t[0].size = 0;
}

void apply(int i, int x) {
    if (i == 0) return;
    t[i].x += x;
    t[i].sum += x * t[i].size;
    t[i].lazy += x;
}

void flip(int i) {
    if (i == 0) return;
    swap(t[i].l, t[i].r);
    t[i].rev ^= 1;
}

void push(int i) {
    if (i == 0) return;
    if (t[i].rev != 0) {
        flip(t[i].l);
        flip(t[i].r);
        t[i].rev = 0;
    }
    if (t[i].lazy != 0) {
        apply(t[i].l, t[i].lazy);
        apply(t[i].r, t[i].lazy);
        t[i].lazy = 0;
    }
}

void pull(int i) {
    if (i == 0) return;
    t[i].size = t[t[i].l].size + 1 + t[t[i].r].size;
    t[i].sum = t[t[i].l].sum + t[i].x + t[t[i].r].sum;
}

pair<int, int> split(int i, int k) {
    if (i == 0) return {0, 0};
    push(i);
    int s = t[t[i].l].size;
    if (s >= k) {
        auto [l, r] = split(t[i].l, k);
        t[i].l = r;
        pull(i);
        return {l, i};
    } else {
        auto [l, r] = split(t[i].r, k - s - 1);
        t[i].r = l;
        pull(i);
        return {i, r};
    }
}

int merge(int l, int r) {
    if (l == 0 or r == 0) return l | r;
    if (t[l].p > t[r].p) {
        push(l);
        t[l].r = merge(t[l].r, r);
        pull(l);
        return l;
    } else {
        push(r);
        t[r].l = merge(l, t[r].l);
        pull(r);
        return r;
    }
}

void insert(int k, int x) {
    auto [l, r] = split(root, k);
    int i = t.size();
    t.push_back(node(x));
    root = merge(merge(l, i), r);
}

void erase(int k) {
    auto [l, m] = split(root, k);
    auto [_, r] = split(m, 1);
    root = merge(l, r);
}

void move(int i, int j, int k) {
    auto [h, q] = split(root, j + 1);
    auto [p, m] = split(h, i);
    auto [l, r] = split(merge(p, q), k);
    root = merge(merge(l, m), r);
}

void reverse(int i, int j) {
    auto [h, r] = split(root, j + 1);
    auto [l, m] = split(h, i);
    flip(m);
    root = merge(merge(l, m), r);
}

void update(int i, int j, int x) {
    auto [h, r] = split(root, j + 1);
    auto [l, m] = split(h, i);
    apply(m, x);
    root = merge(merge(l, m), r);
}

int query(int i, int j) {
    auto [h, r] = split(root, j + 1);
    auto [l, m] = split(h, i);
    int q = t[m].sum;
    root = merge(merge(l, m), r);
    return q;
}

int kth(int k) {
    int i = root;
    while (i != 0) {
        push(i);
        int s = t[t[i].l].size;
        if (k < s) {
            i = t[i].l;
        } else if (k > s) {
            k -= s + 1;
            i = t[i].r;
        } else {
            return t[i].x;
        }
    }
    return -1;
}
};
```

10.33 treap

```
// Treap: randomized binary search tree (split/merge)
// Time: operations O(log N) average | Space: O(N)
// Note: combines bst and heap properties | requires mt19937 for
    priorities
struct treap {
    struct node {
        int x, p, l = 0, r = 0, s = 1;

        node(int x, int p = rng()): x(x), p(p) {}
    };

    int root = 0;
    vector<node> t;

    treap(int n = 0) {
        t.reserve(n + 1);
        t.push_back(node(0));
        t[0].s = 0;
    }

    void pull(int i) {
        if (i == 0) return;
        t[i].s = 1 + t[t[i].l].s + t[t[i].r].s;
    }

    pair<int, int> split(int i, int x) {
        if (i == 0) return {0, 0};
        if (t[i].x <= x) {
            auto [l, r] = split(t[i].r, x);
            t[i].r = l;
            pull(i);
            return {i, r};
        } else {
            auto [l, r] = split(t[i].l, x);
            t[i].l = r;
            pull(i);
            return {l, i};
        }
    }

    int merge(int l, int r) {
        if (l == 0 or r == 0) return l | r;
        if (t[l].p > t[r].p) {
            t[l].r = merge(t[l].r, r);
            pull(l);
            return l;
        } else {
            t[r].l = merge(l, t[r].l);
            pull(r);
            return r;
        }
    }

    void insert(int x) {
        auto [l, r] = split(root, x);
        int i = t.size();
        t.push_back(node(x));
        root = merge(merge(l, i), r);
    }

    void erase(int x) {
        auto [l, h] = split(root, x - 1);
        auto [m, r] = split(h, x);
        m = merge(t[m].l, t[m].r);
        root = merge(merge(l, m), r);
    }

    int index(int x) {
        int i = root, k = 0;
        while (i != 0) {
            if (t[i].x < x) k += 1 + t[t[i].l].s, i = t[i].r;
            else i = t[i].l;
        }
        return k;
    }

    int kth(int k) {
        int i = root;
        while (i != 0) {
            int s = t[t[i].l].s;
            if (k < s) i = t[i].l;
            else if (k > s) k -= 1 + s, i = t[i].r;
            else return t[i].x;
        }
        return -1;
    }
};
```

10.34 wavelet-matrix

```
// Wavelet Matrix: range quantile, range frequency, k-th smallest
// Time: build O(N log V) | Queries O(log V) | Space: O(N log V)
// Note: coordinate compression is REQUIRED before building.
struct wavelet_matrix {
    int n, log;
    vector<vector<int>> pref;
    vector<int> mid;

    wavelet_matrix(vector<int> a): n(a.size()),
         log(__lg(*max_element(a.begin(), a.end()) | 1)), pref(log + 1
         , vector<int>(n + 1)), mid(log + 1) {
        for (int b = log; b >= 0; --b) {
            for (int i = 0; i < n; ++i) pref[b][i + 1] = pref[b][i] +
                 (((a[i] >> b) & 1) == 0);
            mid[b] = stable_partition(a.begin(), a.end(), [&](int x)
                 {return ((x >> b) & 1) == 0;}) - a.begin();
        }
    }

    int kth(int i, int j, int k) {
        int x = 0;
        for (int b = log; b >= 0; --b) {
            int l = pref[b][i], r = pref[b][j + 1], z = r - l;
            if (k < z) i = l, j = r - 1;
            else x |= 1 << b, i += mid[b] - l, j += mid[b] - r, k -=
                 z;
        }
        return x;
    }

    int less_or_equal(int i, int j, int x) {
        int c = 0;
        if (x >= (1 << (log + 1))) x = (1 << (log + 1)) - 1;
        for (int b = log; b >= 0; --b) {
            int l = pref[b][i], r = pref[b][j + 1], z = r - l;
            if (((x >> b) & 1) == 0) i = l, j = r - 1;
            else c += z, i += mid[b] - l, j += mid[b] - r;
        }
        if (i <= j) c += j - i + 1;
        return c;
    }
};
```

11.1 burnside-lemma

Lema de Burnside

O Lema de Burnside é utilizado em combinatória matemática para contar o número de objetos estruturalmente únicos (classes de equivalência) em um conjunto, eliminando duplicatas geradas por simetrias (como rotações ou reflexões).

Fórmula Geral:

$$|X/G| = \frac{1}{|G|} \sum_{\pi \in G} |X^\pi|$$

Onde:

- $|X/G|$: Número de objetos únicos.
- $|G|$: Tamanho do grupo de simetrias (total de permutações).
- $|X^\pi|$: Número de pontos fixos (representações que não mudam ao aplicar a permutação π).

Aplicação Clássica: Problema do Colar (Necklace)

Conta o número de formas únicas de montar um colar de tamanho n com m cores possíveis para as pérolas, considerando que colares rotacionados são iguais.
Para uma rotação de tamanho i em um colar de tamanho n, a permutação se decompõe em $\gcd(n, i)$ ciclos. Para que a coloração seja um ponto fixo sob essa rotação, todas as pérolas de um mesmo ciclo devem ter a mesma cor. Como temos m cores, existem $m^{\gcd(n,i)}$ colorações invariantes (pontos fixos) para essa rotação.

Fórmula para Rotações em O(N):

$$\text{Colares} = \frac{1}{n} \sum_{i=1}^{n} m^{\gcd(n,i)} \pmod{10^9 + 7}$$

11.2 exponent-reduction-euler-theorem

Redução de Expoentes Grandes (Teorema Estendido de Euler)

Para calcular $a^e \pmod m$ quando o expoente e é extremamente grande (como em strings com 10^5 dígitos ou torres de potências a^{b^c}), utiliza-se a seguinte identidade fundamental:

$$a^e \equiv \begin{cases} a^e \pmod m, & \text{se } e < \varphi(m) \\ a^{(e \bmod \varphi(m)) + \varphi(m)} \pmod m, & \text{se } e \geq \varphi(m) \end{cases}$$

Resumo de Aplicação

- **Quando se aplica:** Funciona universalmente para **qualquer** par de inteiros a e m, mesmo quando a e m não são primos entre si ($\gcd(a, m) \neq 1$).
- **Caso** $e < \varphi(m)$**:** Não reduza o expoente. Reduzi-lo antes de atingir $\varphi(m)$ altera o resultado quando $\gcd(a, m) > 1$.
- **Caso** $e \geq \varphi(m)$**:** Substitua o expoente e por $(e \bmod \varphi(m)) + \varphi(m)$. A adição do termo $+\varphi(m)$ garante que a fase transitória da potenciação seja respeitada.

11.3 flow-bipartite-graphs-reductions

Fluxo, Grafos Bipartidos e Reduções

Equivalências em Grafos Bipartidos

Dado um grafo bipartido $G = (L \cup R, E)$ e o Pareamento Máximo M (obtido via Fluxo Máximo $S \to L \to R \to T$ com capacidades 1):

- **Max Bipartite Matching (MBM):** Valor do fluxo máximo $|f_{\max}|$.
- **Min Vertex Cover (MVC):** Mínimo de vértices para cobrir todas as arestas.

$$|\text{MVC}| = |\text{MBM}| \quad \text{(Teorema de König)}$$

- **Max Independent Set (MIS):** Máximo de vértices sem arestas entre si.

$$|\text{MIS}| = |V| - |\text{MVC}| = |V| - |\text{MBM}|$$

- **Min Path Cover (DAG):** Menor número de caminhos vértices-disjuntos para cobrir um DAG. Duplique cada vértice $u \to (u_L, u_R)$. Para cada aresta $u \to v$, adicione (u_L, v_R) com capacidade 1.

$$\text{Quantidade de Caminhos} = |V| - |\text{MBM}|$$

- **Teorema de Dilworth (Anticadeias em DAGs):** Uma anticadeia é o maior conjunto de vértices onde nenhum alcança o outro.

$$\text{Tamanho da Maior Anticadeia} = \text{Quantidade Mínima de Caminhos}$$

Teoremas em Grafos Gerais (Não Bipartidos)

Enquanto o Teorema de König falha fora de grafos bipartidos, as **Identidades de Gallai** valem para **qualquer grafo** $G = (V, E)$ sem vértices isolados:

- **Identidades de Gallai:**

$$|\text{MIS}| + |\text{MVC}| = |V|$$

$$|\text{Max Matching}| + |\text{Min Edge Cover}| = |V|$$

- **Min Edge Cover (Cobertura Mínima de Arestas):** Menor número de arestas necessárias para cobrir todos os vértices.

$$|\text{Min Edge Cover}| = |V| - |\text{Max Matching}|$$

Como obter: Encontre o Pareamento Máximo. Para cada vértice não pareado, adicione qualquer aresta incidente a ele.

- **Alertas para Programação Competitiva:**
  - **König não vale:** $|\text{MVC}| \neq |\text{Max Matching}|$ em grafos gerais (ex: no triângulo C_3, $|\text{MVC}| = 2$ e $|\text{Matching}| = 1$).
  - **Complexidade:** MVC e MIS tornam-se problemas **NP-Hard** em grafos gerais.
  - **Fluxo não resolve Pareamento Geral:** Para pareamento em grafos arbitrários (com ciclos ímpares), o algoritmo de fluxo falha. É necessário usar o **Algoritmo de Blossom (Edmonds)**.

Como Recuperar a Resposta (Reconstrução)

Passo Inicial (Grafo Residual G_f): Rode uma DFS ou BFS a partir da fonte S considerando apenas arestas com **capacidade residual** > 0.
Guarde os vértices visitados no conjunto S_{reach}.

- **Pareamento Máximo (MBM):**
  - Uma aresta original (u, v) com $u \in L$ e $v \in R$ pertence ao pareamento se tiver **fluxo positivo**.
- **Min Cut (Arestas do Corte Mínimo):**
  - O corte divide o grafo em visitados (S_{reach}) e não visitados (T_{reach}).
  - Uma aresta (u, v) do **grafo original** pertence ao corte se u foi visitado e v não foi visitado:

$$u \in S_{\text{reach}} \quad \text{e} \quad v \notin S_{\text{reach}}$$

  - Essas arestas estarão obrigatoriamente **saturadas** (fluxo = capacidade).
- **Min Vertex Cover (MVC em Bipartidos):**
  - Pegue os vértices do lado esquerdo L **NÃO visitados** pela DFS.
  - Pegue os vértices do lado direito R **VISITADOS** pela DFS.

$$\text{MVC} = (L \setminus S_{\text{reach}}) \cup (R \cap S_{\text{reach}})$$

- **Max Independent Set (MIS em Bipartidos):**
  - É o **complemento** do Min Vertex Cover ($V \setminus \text{MVC}$):

$$\text{MIS} = (L \cap S_{\text{reach}}) \cup (R \setminus S_{\text{reach}})$$

  - Selecione L **VISITADOS** e R **NÃO visitados**.
- **Min Path Cover (Recuperar Caminhos no DAG):**
  - Reconstrua as cadeias seguindo as arestas (u_L, v_R) que possuem fluxo = 1.
  - Todo vértice u_L que **não** tem fluxo saindo representa o **fim** de um caminho.

Reduções Clássicas de Fluxo

- **Project Selection (Maximum Weight Closure):** Queremos escolher um subconjunto de itens para maximizar o ganho líquido. Se um item u for escolhido e depender de v ($u \to v$), o item v obrigatoriamente precisa ser escolhido junto.
  - Item com **lucro** $p_u > 0$: aresta $S \to u$ com capacidade p_u.
  - Item com **custo** $c_u > 0$: aresta $u \to T$ com capacidade c_u.
  - **Dependência** $u \to v$: aresta $u \to v$ com capacidade ∞.
  - **Corte Mínimo:** Representa a perda mínima.
  - **Lucro Máximo:** $\sum$(Lucros Positivos) − Min Cut.
  - **Itens Selecionados:** Conjunto $S_{\text{reach}} \setminus \{S\}$.
- **Capacidade em Vértices:** Split no vértice $u \to (u_{\text{in}}, u_{\text{out}})$. Adicione aresta $u_{\text{in}} \to u_{\text{out}}$ com cap $= c(u)$. Arestas que entram em u vão para u_{in}, e as que saem vão de u_{out}.

Fluxo com Limite Inferior nas Arestas (Circulation with Demands)

Use essa técnica quando uma aresta $u \to v$ possuir uma capacidade mínima obrigatória L e uma máxima R ($L \leq f \leq R$). O algoritmo tradicional não entende limites inferiores, então precisamos usar vértices fictícios para forçar esse fluxo.
Construção da Rede:

- Crie dois vértices auxiliares: uma **Super Fonte** (SS) e um **Super Sumidouro** (TT).
- Para cada aresta original $u \to v$ com limite $[L, R]$, adicione na nova rede uma aresta $u \to v$ com **capacidade** $R - L$.
- Mantenha um balanço para cada vértice v. Seja $in[v]$ a soma de todos os limites L das arestas que **entram** em v, e $out[v]$ a soma de todos os limites L das arestas que **saem** de v.
- Adicione uma aresta $SS \to v$ com capacidade $in[v]$.
- Adicione uma aresta $v \to TT$ com capacidade $out[v]$.
- Adicione uma aresta do sumidouro original para a fonte original ($T \to S$) com capacidade ∞ (para permitir a circulação).

Verificação e Resposta:

- Rode o Fluxo Máximo partindo de SS para TT.
- **Existe solução válida?** Apenas se **todas** as arestas saindo de SS (ou chegando em TT) estiverem totalmente saturadas. Ou seja, se o Max Flow $= \sum in[v]$.
- **Recuperar fluxo nas arestas:** O fluxo real em uma aresta original $u \to v$ será: L + (fluxo que passou pela aresta $u \to v$ na nova rede).
- **Para achar o Max Flow real de** S **a** T**:** Após verificar que o fluxo inicial é válido, remova os vértices SS e TT e suas arestas, remova a aresta infinita $T \to S$, e rode mais uma vez o Fluxo Máximo de S para T na rede residual.

11.4 johnson-algorithm-potential-method

O Algoritmo de Johnson e o Método dos Potenciais

Na programação competitiva, o problema de Caminhos Mais Curtos entre Todos os Pares (APSP - *All-Pairs Shortest Paths*) é frequentemente resolvido com o algoritmo de Floyd-Warshall em $\mathcal{O}(V^3)$. No entanto, para grafos esparsos, essa complexidade pode ser um gargalo. O Algoritmo de Johnson resolve o APSP em $\mathcal{O}(VE \log V)$, o que é muito mais eficiente em grafos onde $E \ll V^2$, mesmo na presença de arestas com pesos negativos.

O Método dos Potenciais

Se o grafo tiver apenas pesos não-negativos, poderíamos simplesmente executar o algoritmo de Dijkstra a partir de cada vértice, obtendo $\mathcal{O}(VE \log V)$. O problema surge com arestas negativas, que quebram a premissa gulosa do Dijkstra.
Para contornar isso, usamos o **Método dos Potenciais**. A ideia é atribuir um valor (potencial) $h(u)$ a cada vértice $u \in V$ e reponderar cada aresta (u, v) de peso $w(u, v)$ com um novo peso $w'(u, v)$ definido como:

$$w'(u, v) = w(u, v) + h(u) - h(v)$$

Essa reponderação tem duas propriedades matemáticas fundamentais:

1. **Preservação dos Caminhos Mais Curtos:** Para qualquer caminho $p = v_0 \to v_1 \to \cdots \to v_k$, o novo peso total do caminho é:

   $$w'(p) = w(p) + h(v_0) - h(v_k)$$

   Como o termo $h(v_0) - h(v_k)$ depende apenas dos vértices de início e fim, todos os caminhos entre v_0 e v_k sofrem exatamente a mesma alteração. Logo, o caminho mais curto no grafo original continua sendo o caminho mais curto no grafo reponderado.
2. **Ciclos Negativos:** O peso de um ciclo não é alterado pela reponderação, pois o vértice de início e fim é o mesmo, o que faz com que $h(v_0) - h(v_k) = 0$.

O que torna um Potencial Válido?

Nosso objetivo ao alterar os pesos é habilitar o uso do algoritmo de Dijkstra, que exige que todas as arestas possuam peso não-negativo para garantir a sua corretude. Assim, um potencial h é considerado **válido** se, e somente se, para toda aresta $(u, v) \in E$, o novo peso não for negativo:

$$w'(u, v) \geq 0$$

Substituindo a definição de w', isso significa que:

$$w(u, v) + h(u) - h(v) \geq 0$$

Isolando os potenciais, chegamos à restrição fundamental de um potencial válido:

$$h(v) - h(u) \leq w(u, v)$$

Essa inequação tem exatamente o mesmo formato da desigualdade triangular que rege os caminhos mais curtos em grafos.
Isso também nos prova matematicamente por que é impossível resolver o problema se houver um **ciclo de peso negativo**. Imagine um ciclo $C = v_1 \to v_2 \to \cdots \to v_k \to v_1$. Se tentarmos encontrar um potencial válido e somarmos a inequação $h(v) - h(u) \leq w(u, v)$ para todas as arestas desse ciclo, o lado esquerdo será exatamente 0 (pois todo vértice aparece uma vez subtraindo e uma vez somando). O lado direito será a soma dos pesos do ciclo, $w(C)$. Teríamos então:

$$0 \leq w(C)$$

Se $w(C)$ for um valor negativo, temos um absurdo matemático ($0 \leq$ negativo). Isso prova de forma elegante que, se o grafo possui um ciclo de peso negativo, é matematicamente impossível formular uma função potencial válida.

O Algoritmo de Johnson

O objetivo de Johnson é encontrar essa função potencial h válida. Se garantirmos isso, podemos rodar o Dijkstra em segurança a partir de qualquer vértice.
Os passos do algoritmo são:

1. **Vértice Virtual:** Adicione um novo vértice s ao grafo, com arestas de peso 0 direcionadas para todos os outros vértices do grafo original.
2. **Bellman-Ford:** Execute o algoritmo de Bellman-Ford a partir do vértice s.
   - Se o Bellman-Ford detectar um ciclo de peso negativo, o algoritmo para, pois o problema de caminho mais curto não tem solução.
   - Caso contrário, defina o potencial de cada vértice u como a distância mínima de s até u, ou seja, $h(u) = \delta(s, u)$.
3. **Reponderação:** Pela desigualdade triangular do caminho mais curto, sabemos que $\delta(s, v) \leq \delta(s, u) + w(u, v)$. Substituindo por h, temos $h(v) \leq h(u) + w(u, v)$. Rearranjando a inequação, obtemos:

   $$w(u, v) + h(u) - h(v) \geq 0$$

   Logo, definimos $w'(u, v) = w(u, v) + h(u) - h(v)$ e agora garantimos que todas as arestas são não-negativas.
4. **Dijkstra Múltiplo:** Remova o vértice virtual s. Para cada vértice $u \in V$, execute o algoritmo de Dijkstra no grafo reponderado para encontrar as menores distâncias $d'(u, v)$.
5. **Reajuste das Distâncias:** A distância real no grafo original é recuperada revertendo a fórmula do potencial:

   $$d(u, v) = d'(u, v) - h(u) + h(v)$$

Na prática das maratonas de programação, o Algoritmo de Johnson é a ferramenta ideal quando você se depara com um grafo grande e esparso, contendo arestas de peso negativo, e que exigiria múltiplas consultas, situação em que o algoritmo de Floyd-Warshall resultaria inevitavelmente em *Time Limit Exceeded* (TLE).

11.5 manhattan-to-chebyshev

Manhattan para Chebyshev

Na programação competitiva, problemas que envolvem distâncias em um grid (plano 2D) frequentemente utilizam a **Distância de Manhattan**. A distância de Manhattan entre dois pontos $A = (x_1, y_1)$ e $B = (x_2, y_2)$ é calculada pela soma das diferenças absolutas de suas coordenadas:

$$d_M(A, B) = |x_1 - x_2| + |y_1 - y_2|$$

A Dificuldade da Distância de Manhattan

Matematicamente, lidar com a soma de dois valores absolutos pode ser desconfortável e custoso. Como as coordenadas x e y estão "amarradas" na mesma equação pela soma, não podemos analisá-las ou otimizá-las separadamente.
Se um problema pede para encontrarmos a maior distância entre N pontos, testar todos os pares possíveis exigiria um tempo da ordem de $O(N^2)$, o que é muito lento e frequentemente resulta em *Time Limit Exceeded* (TLE) em competições.
Precisamos de uma forma de separar x e y.

A Transformação: Rotacionando o Plano

Para desvincular as coordenadas, utilizamos um truque geométrico elegante: rotacionar o sistema de coordenadas em 45 graus. Na prática, pegamos cada ponto original (x, y) e o convertemos em um novo ponto (u, v) aplicando as seguintes equações:

$$u = x + y$$
$$v = x - y$$

Neste novo plano de coordenadas (u, v), medimos a distância usando a **Distância de Chebyshev**. Em vez de somar as diferenças, a distância de Chebyshev pega o valor máximo entre as diferenças dos eixos:

$$d_C(A', B') = \max(|u_1 - u_2|, |v_1 - v_2|)$$

A grande "mágica" dessa transformação é a garantia de que a distância de Manhattan no plano original é **exatamente igual** à distância de Chebyshev no plano transformado:

$$d_M(A, B) = d_C(A', B')$$

A Intuição: Por que isso é tão útil?

A vantagem monumental de converter a fórmula para usar a função max é a **independência das dimensões**.
Na fórmula original da soma, um x pequeno poderia ser compensado por um y grande, nos obrigando a olhar para o par sempre junto. Com a função max, podemos olhar para o eixo u isoladamente e para o eixo v isoladamente.

Voltando ao exemplo de encontrar a distância máxima entre N pontos: com o plano transformado, tudo o que precisamos fazer é encontrar a maior distância no eixo u (ou seja, o maior u menos o menor u) e a maior distância no eixo v (o maior v menos o menor v). O resultado final será simplesmente o maior desses dois valores. Isso derruba a complexidade do problema de $O(N^2)$ para um rápido e eficiente $O(N)$.

11.6 mobius-inversion

Inversão de Möbius

A Inversão de Möbius é uma técnica poderosa na teoria dos números e combinatória, frequentemente utilizada para inverter somatórios sobre divisores ou para calcular a cardinalidade de conjuntos com propriedades de coprimalidade restritas.
Fórmula Geral (Forma de Divisores):

$$g(n) = \sum_{d|n} f(d) \iff f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right)$$

Onde $\mu(d)$ é a função de Möbius, definida como:

- $\mu(1) = 1$.
- $\mu(d) = 0$ se d possui um fator quadrado perfeito (i.e., $p^2 \mid d$).
- $\mu(d) = (-1)^k$ se d é o produto de k números primos distintos.

A propriedade fundamental que conecta essa função à contagem de MDC é dada por:

$$\sum_{d|\gcd(a,b)} \mu(d) = \begin{cases} 1, & \text{se } \gcd(a, b) = 1 \\ 0, & \text{se } \gcd(a, b) > 1 \end{cases}$$

Aplicação Clássica: Pares com MDC igual a x

Dado um vetor A de tamanho n com elementos positivos, o objetivo é contar quantos pares de índices (i, j) com $1 \leq i, j \leq n$ satisfazem $\gcd(A[i], A[j]) = x$.
Para resolver esse problema, definimos uma função auxiliar $C(v)$, que representa a quantidade de elementos no vetor A que são múltiplos de v.
Dedução da Fórmula: Queremos contar os pares (i, j) tais que $\gcd(A[i], A[j]) = x$. Isso equivale a exigir que $A[i]$ e $A[j]$ sejam múltiplos de x, e que ao dividi-los por x, eles sejam coprimos: $\gcd\left(\frac{A[i]}{x}, \frac{A[j]}{x}\right) = 1$.
Podemos escrever a quantidade total de pares desejados como:

$$\text{Pares} = \sum_{\substack{1 \leq i \leq n \\ x | A[i]}} \sum_{\substack{1 \leq j \leq n \\ x | A[j]}} \left[\gcd\left(\frac{A[i]}{x}, \frac{A[j]}{x}\right) = 1\right]$$

Substituindo a condição de coprimalidade pela propriedade fundamental de Möbius, temos:

$$\text{Pares} = \sum_{\substack{1 \leq i \leq n \\ x | A[i]}} \sum_{\substack{1 \leq j \leq n \\ x | A[j]}} \sum_{k | \gcd\left(\frac{A[i]}{x}, \frac{A[j]}{x}\right)} \mu(k)$$

A condição de que k divide o gcd de $\frac{A[i]}{x}$ e $\frac{A[j]}{x}$ significa que k divide $\frac{A[i]}{x}$ e k divide $\frac{A[j]}{x}$. Isso é matematicamente equivalente a dizer que $(k \cdot x)$ divide tanto $A[i]$ quanto $A[j]$.
Agora, invertemos a ordem dos somatórios. Em vez de iterar sobre os pares e depois sobre os divisores, iteramos sobre k (o multiplicador) e contamos quantos pares satisfazem a condição de serem múltiplos de $(k \cdot x)$.
Como o número de elementos em A que são múltiplos de $(k \cdot x)$ é $C(k \cdot x)$, o número de pares que podemos formar com esses elementos é simplesmente $C(k \cdot x)^2$. Agrupando tudo, chegamos à fórmula final.
Fórmula para Contagem de Pares:

$$\text{Pares} = \sum_{k=1}^{\lfloor M/x \rfloor} \mu(k) \cdot C(k \cdot x)^2 \pmod{10^9 + 7}$$

Onde M é o valor máximo presente no vetor A. A função $C(v)$ pode ser pré-computada para todos os valores em $O(M \log M)$ através de uma abordagem semelhante ao Crivo de Eratóstenes, permitindo responder à consulta para qualquer x em $O(M/x)$.

11.7 vandermonde-identity

Identidade de Vandermonde

A Identidade de Vandermonde é uma ferramenta essencial em combinatória para otimizar somatórios envolvendo multiplicações de coeficientes binomiais, reduzindo cálculos de complexidade $O(N)$ para $O(1)$ (com pré-computação de fatoriais).

Fórmula Clássica

Para quaisquer inteiros não negativos m, n e r:

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$$

Intuição Combinatória: Escolher um comitê de r pessoas de um grupo contendo m homens e n mulheres é equivalente a somar todos os cenários possíveis de escolha (onde escolhemos k homens e $r-k$ mulheres).

Uso em Programação Competitiva (O Truque da Simetria)

Em problemas de contagem, raramente a fórmula aparece pronta. O cenário mais comum é encontrar uma soma onde os índices inferiores **crescem simultaneamente**.

Para aplicar Vandermonde, a soma dos termos inferiores deve ser **constante**. Alcançamos isso aplicando a simetria do binômio:

$$\binom{n}{k} = \binom{n}{n-k}$$

Padrão 1: Índices Iguais

Se a sua modelagem resultou na soma:

$$\sum_{k=0}^{\min(x,y)} \binom{x}{k}\binom{y}{k}$$

Aplique a simetria no primeiro termo ($\binom{x}{k} \to \binom{x}{x-k}$):

$$\sum_{k} \binom{x}{x-k}\binom{y}{k} = \binom{x+y}{x}$$

Nota: A soma dos índices inferiores agora é $(x-k)+k = x$ (constante).

Padrão 2: Índices Deslocados (Exemplo Prático)

Se você precisar escolher $k-1$ itens de um grupo e k de outro:

$$\sum_{k\geq 1} \binom{x}{k-1}\binom{y}{k}$$

Transforme o primeiro termo usando $\binom{x}{k-1} = \binom{x}{x-(k-1)} = \binom{x}{x-k+1}$:

$$\sum_{k\geq 1} \binom{x}{x-k+1}\binom{y}{k} = \binom{x+y}{x+1}$$

Dica de Implementação: Sempre pré-compute os fatoriais e seus inversos modulares para responder a $\binom{n}{k} \pmod M$ em $O(1)$.